

Foreword

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where:

x the first digit:

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y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

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1 Scope

The present document establishes the minimum RF requirements for NR User Equipment (UE) Interworking operation with other radios. This includes but is not limited to additional requirements for carrier aggregation or NR dual connectivity between Range 1 and Range 2 and additional requirements due to NR non-standalone (NSA) operation mode with E-UTRA.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or nonspecific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

- [1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".
- [2] 3GPP TS 38.101-1: "NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone"
- [3] 3GPP TS 38.101-2: "NR; User Equipment (UE) radio transmission and reception; Part 2: Range 2 Standalone"
- [4] 3GPP TS 36.101: "Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) radio transmission and reception"
- [5] 3GPP TS 38.521-3: "NR; User Equipment (UE) conformance specification; Radio transmission and reception; Part 3: Range 1 and Range 2 Interworking operation with other radios"
- [6] Recommendation ITU-R M.1545: "Measurement uncertainty as it applies to test limits for the terrestrial component of International Mobile Telecommunications-2000"
- [7] 3GPP TS 36.211: "E-UTRA; Physical channels and modulation"
- [8] 3GPP TS 36.331: " Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification"
- [9] 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- [10] 3GPP TS 38.213: "NR; Physical layer procedures for control"
- [11] 3GPP TS 38.306: "NR; User Equipment (UE) radio access capabilities"
- [12] 3GPP TS 38.133: "NR; Requirements for support of radio resource management".
- [13] 3GPP TS 38.211: "NR; Physical channels and modulation".

3 Definitions, symbols and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

3.2 Symbols

For the purposes of the present document, the following symbols apply:

$\Delta R_{IB,c}$	Allowed reference sensitivity relaxation due to support for CA or DC operation, for serving cell c .
$\Delta T_{IB,c}$	Allowed maximum configured output power relaxation due to support for CA or DC operation, for serving cell c
$BW_{E-UTRA_Channel}$	Channel bandwidth of E-UTRA carrier
$BW_{E-UTRA_Channel_CA}$	Channel bandwidth of E-UTRA sub-block which is composed of intra-band contiguous CA E-UTRA carriers
$BW_{NR_Channel}$	Channel bandwidth of NR carrier
$BW_{NR_Channel_CA}$	Channel bandwidth of NR sub-block which is composed of intra-band contiguous CA NR carriers
$Ceil(x)$	Rounding upwards; $ceil(x)$ is the smallest integer such that $ceil(x) \geq x$
$EN-DC_{ACLR}$	The ratio of the filtered mean power centred on the aggregated sub-block bandwidth ENBW to the filtered mean power centred on an adjacent bandwidth of the same size ENBW
$E-UTRA_{ACLR}$	E-UTRA ACLR
F_C	<i>RF reference frequency</i> for the carrier center on the channel raster
F_{DL_low}	The lowest frequency of the downlink <i>operating band</i>
F_{DL_high}	The highest frequency of the downlink <i>operating band</i>
F_{UL_low}	The lowest frequency of the uplink <i>operating band</i>
F_{UL_high}	The highest frequency of the uplink <i>operating band</i>
F_{OOB}	The boundary between the NR out of band emission and spurious emission domains
L_{CRB}	Transmission bandwidth which represents the length of a contiguous resource block allocation expressed in units of resource blocks
$Max()$	The largest of given numbers
$Min()$	The smallest of given numbers
NR_{ACLR}	NR ACLR
N_{RB}	Transmission bandwidth configuration, expressed in units of resource blocks
N_{RB_agg}	The number of the aggregated RBs within the fully allocated aggregated channel bandwidth for carrier 1 to j , where μ is defined in TS 38.211 [13]
$N_{RB,c}$	The transmission bandwidth configuration of component carrier c , expressed in units of resource blocks for carrier j , where μ is defined in TS 38.211 [13]
P_{CMAX}	The configured maximum UE output power
RB_{start}	Indicates the lowest RB index of transmitted resource blocks
W_{gap}	The sub-block gap between the two sub-blocks

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

ACLR	Adjacent Channel Leakage Ratio
ACS	Adjacent Channel Selectivity
A-MPR	Additional Maximum Power Reduction
BCS	Bandwidth Combination Set
CA	Carrier Aggregation
CC	Component Carrier
DC	Dual Connectivity
EN-DC	E-UTRA/NR DC
EVM	Error Vector Magnitude
FDM	Frequency Division Multiplexing
FR	Frequency Range
ENBW	The aggregated bandwidth of an E-UTRA sub-block and an adjacent NR sub-block
ITU-R	Radiocommunication Sector of the International Telecommunication Union
MBW	Measurement bandwidth defined for the protected band
MPR	Allowed maximum power reduction
MSD	Maximum Sensitivity Degradation
MCG	Master Cell Group
NR	New Radio
NS	Network Signalling
NSA	Non-Standalone, a mode of operation where operation of an other radio is assisted with an other radio
OOB	Out-of-band
OOBE	Out-of-band emission
OTA	Over The Air
PRB	Physical Resource Block
RE	Resource Element
REFSENS	Reference Sensitivity
RF	Radio Frequency
Rx	Receiver
SCG	Secondary Cell Group
SCS	Subcarrier spacing
SEM	Spectrum Emission Mask
SUL	Supplementary uplink
TDM	Time Division Multiplex
Tx	Transmitter
UE	User Equipment
UL MIMO	Up Link Multiple Antenna transmission
ULSUP	Uplink sharing from UE perspective

4 General

4.1 Relationship between minimum requirements and test requirements

The present document is interwork specification for NR UE, covering RF characteristics and minimum performance requirements. Conformance to the present specification is demonstrated by fulfilling the test requirements specified in the conformance specification 3GPP TS 38.521-3 [5].

The Minimum Requirements given in this specification make no allowance for measurement uncertainty. The test specification TS 38.521-3 [5] defines test tolerances. These test tolerances are individually calculated for each test. The test tolerances are used to relax the minimum requirements in this specification to create test requirements. For some requirements, including regulatory requirements, the test tolerance is set to zero.

The measurement results returned by the test system are compared - without any modification - against the test requirements as defined by the shared risk principle. The shared risk principle is defined in Recommendation ITUR M.1545 [6].

4.2 Applicability of minimum requirements

- a) In this specification the Minimum Requirements are specified as general requirements and additional requirements. Where the Requirement is specified as a general requirement, the requirement is mandated to be met in all scenarios
- b) For specific scenarios for which an additional requirement is specified, in addition to meeting the general requirement, the UE is mandated to meet the additional requirements.
- c) The spurious emissions power requirements are for the long-term average of the power. For the purpose of reducing measurement uncertainty it is acceptable to average the measured power over a period of time sufficient to reduce the uncertainty due to the statistical nature of the signal
- d) Terminal that supports EN-DC configuration shall meet E-UTRA requirements as specified in TS 36.101 [4] and NR requirements as in TS 38.101-1 [2] and TS 38.101-2 [3] unless otherwise specified in this specification
- e) All the requirements for intra-band contiguous and non-contiguous EN-DC apply under the assumption of the same uplink-downlink and special subframe configurations in the E-UTRA and slot format indicated by UL-DL-configurationCommon and UL-DL-configurationDedicated in the NR for the EN-DC, a time offset between the two RATs configurations may be required.
- f) For EN-DC combinations with CA configurations for E-UTRA and/or NR, all the requirements for E-UTRA and/or NR all the requirements for E-UTRA and/or NR intra-band contiguous and non-contiguous CA apply under the assumption of the same slot format indicated by UL-DL-configurationCommon and UL-DL-configurationDedicated in the PCell and SCells for NR and the same uplink-downlink and special subframe configurations in Pcell and SCells for E-UTRA.

A terminal which supports an EN-DC configuration shall support:

If any subsets of the EN-DC configuration do not specify its own bandwidth combination sets in 5.3B, then the terminal shall support the same E-UTRA bandwidth combination sets it signals the support for in E-UTRA CA configuration part of E-UTRA – NR DC and shall support the same NR bandwidth combination sets it signals the support for in NR CA configuration part of E-UTRA – NR DC.

Else if one of the subsets of the EN-DC configuration specify its own bandwidth combination sets in 5.3B, then the terminal shall support a product set of channel bandwidth for each band specified by E-UTRA bandwidth combination sets, NR bandwidth combination sets, and EN-DC bandwidth combination sets it signals the support.

A terminal which supports an inter-band EN-DC configuration with a certain UL configuration shall support the all lower order DL configurations of the lower order EN-DC combinations, which have this certain UL configuration and the fallbacks of this UL configuration.

For CA or DC configurations, which include FR2 intra-band CA configurations with multiple FR2 sub-blocks, where at least one of the sub-blocks is a contiguous CA configuration:

- if the field *partialFR2-FallbackRX-Req* is not present, the UE shall meet all applicable UE RF requirements for the highest order CA configuration and all associated fallback CA configurations;
- if the field *partialFR2-FallbackRX-Req* is present, for each FR2 intra-band CA configuration with multiple sub-blocks that the UE indicates support for explicitly in UE capability signalling: the in-gap UE RF requirements in clauses 7.5A, 7.5B, 7.6A, 7.6B apply as the equivalent requirements for the associated fallback FR2 intra-band CA configurations with the same number of sub-blocks, where at least one of the sub-blocks consists of a contiguous CA configuration. The UE shall meet all applicable UE RF requirements for fallback CA configurations with a lesser number of sub-blocks;
- regardless of the field *partialFR2-FallbackRX-Req*, the UE shall meet all DL out-of-gap requirements for all lower order fallback CA configurations.

Terminal that supports inter-band NR-DC between FR1 and FR2 configuration shall meet the requirements for corresponding CA configuration (suffix A), unless otherwise specified.

4.3 Specification suffix information

Unless stated otherwise the following suffixes are used for indicating at 2nd level clause, shown in Table 4.3-1.

Table 4.3-1: Definition of suffixes

Clause suffix	Variant
None	Single Carrier
A	Carrier Aggregation (CA) between FR1 and FR2
B	Dual-Connectivity (DC) with and without SUL including UL sharing from UE perspective, inter-band NR DC between FR1 and FR2
D	UL MIMO

5 Operating bands and channel arrangement

5.1 General

The channel arrangements presented in this clause are based on the operating bands and channel bandwidths defined in the present release of specifications.

NOTE: Other operating bands and channel bandwidths may be considered in future releases.

Requirements throughout the RF specifications are in many cases defined separately for different frequency ranges (FR). The frequency ranges in which NR can operate according to this version of the specifications are identified as described in Table 5.1-1.

Table 5.1-1: Definition of frequency ranges

Frequency range designation	Corresponding frequency range
FR1	410 MHz – 7125 MHz
FR2	24250 MHz – 52600 MHz

The present specification covers band combinations including

- at least one FR1 operating band and one FR2 operating band for carrier aggregation and dual connectivity operations;
- at least one E-UTRA operating band for dual connectivity operations.

5.2 Operating bands

NR is designed to operate in FR1 operating bands defined in TS 38.101-1 [2] and FR2 operating bands defined in TS 38.101-2 [3]. E-UTRA is designed to operate in operating bands defined in TS 36.101 [4].

5.2A Operating bands for CA

5.2A.1 Inter-band CA between FR1 and FR2

NR carrier aggregation are designed to operate in the operating bands defined in Table 5.2A.11. The band combinations include at least one FR1 operating band and one FR2 operating band.

If the mandatory simultaneous Rx/Tx capability applies for a lower order band combination, when the applicable lower order band combination is a band pair in a higher order band combination, the mandatory simultaneous Rx/Tx capability also applies for the band pair in the higher order band combination.

Table 5.2A.1-1: Band combinations for inter-band NR CA between FR1 and FR2

NR CA Band	NR Band
CA_n8-n258 ¹	n8, n258
CA_n71-n257 ¹	n71, n257
CA_n77-n257 ¹	n77, n257
CA_n78-n257 ¹	n78, n257
CA_n79-n257 ¹	n79, n257
NOTE 1: Applicable for UE supporting inter-band carrier aggregation with mandatory simultaneous Rx/Tx capability.	

5.2B Operating bands for DC

5.2B.1 General

The operating bands are specified in clause 5.5B for operation with EN-DC, NGEN-DC, NE-DC or NR-DC configured.

5.2B.2 Void

5.2B.3 Void

5.2B.4 Void

5.2B.5 Void

5.2B.6 Void

5.2B.7 Void

5.3 UE Channel bandwidth

5.3A UE Channel bandwidth for CA

5.3A.1 Inter-band CA between FR1 and FR2

For inter-band NR CA between FR1 and FR2, a carrier aggregation configuration is a combination of operating bands, each supporting a carrier aggregation bandwidth class as specified in clause 5.3A.5 of TS 38.101-1 [2] and clause 5.3A.4 of TS 38.101-2 [3] independently.

5.3B UE Channel bandwidth for DC

5.3B.0 General

For intra-band contiguous EN-DC, the aggregated channel bandwidth is sum of the individual NR and E-UTRA channel bandwidths assuming nominal EN-DC channel with 0 kHz offset spacing as specified in clause 5.4.

$$ENBW = BW_{NR_Channel} + BW_{E-UTRA_Channel}$$

In the case where the NR sub-block and/or the E-UTRA sub-block itself is composed of intra-band contiguous CA carriers, the EN-DC aggregated channel bandwidth is the sum of the aggregated channel bandwidths of the NR and E-UTRA sub-blocks assuming nominal EN-DC channel spacing between the NR sub-block and E-UTRA sub-block.

$$ENBW = BW_{NR_Channel_CA} + BW_{E-UTRA_Channel_CA}$$

Intra-band contiguous EN-DC configurations are defined using intra-band contiguous EN-DC bandwidth class notation DC_(n)Xyz where the first EN-DC bandwidth class letter y indicates the number of contiguous E-UTRA carriers and the second EN-DC bandwidth class letter z indicates the number of contiguous NR carriers for the EN-

DC combination of E-UTRA Band X and NR Band nX. Applicable contiguous intra-band EN-DC bandwidth classes are listed in Table 5. 3B.0-1.

Table 5.3B.0-1: Intra-band contiguous EN-DC bandwidth classes

Intra-band contiguous EN-DC bandwidth class	Number of contiguous CC	
	E-UTRA	NR
AA	1	1
CA	2	1
DA	3	1

5.3B.1 Intra-band EN-DC in FR1

5.3B.1.1 General

The requirements for intra-band EN-DC in this specification are defined for EN-DC configurations with associated bandwidth combination sets.

For each EN-DC configuration, requirements are specified for all bandwidth combinations contained in a *bandwidth combination set*, which is indicated per supported band combination in the UE radio access capability. A UE can indicate support of several bandwidth combination sets per band combination.

5.3B.1.2 BCS for Intra-band contiguous EN-DC

For intra-band contiguous EN-DC, an EN-DC configuration is consisting of an E-UTRA band and a corresponding NR band having the same frequency range which supports an intra-band contiguous EN-DC bandwidth class.

Bandwidth combination sets for intra-band contiguous EN-DC are specified in Table 5.3B.1.2-1.

The EN-DC configurations and bandwidth combination sets in Table 5.3B.1.2-1 also apply to higher order EN-DC combinations that include inter-band and intra-band EN-DC on the downlink and inter-band EN-DC on the uplink. If no BCS is reported in the UE capabilities for an intra-band combination the default is that the UE supports BCS0.

Table 5.3B.1.2-1: EN-DC configurations and bandwidth combination sets defined for intra-band contiguous EN-DC

	E-UTRA	A – NR configuration / Bandwidth combination set				
Downlink EN-DC configuration	Uplink EN-DC configurations	Component carriers in order of increasing carrier frequency			Maximum aggregated bandwidth (MHz)	Bandwidth combination set
		Channel bandwidths for E-UTRA carrier (MHz)	Channel bandwidths for NR carrier (MHz)	Channel bandwidths for E-UTRA carrier (MHz)		
DC_(n)41AA	DC_(n)41AA	20	40, 60, 80,100		120	0
			40, 60, 80,100	20		
		20	40, 50, 60, 80,100		120	1
			40, 50, 60, 80,100	20		
DC_(n)41CA	DC_(n)41AA	20+20	40, 60, 80,100		140	0
			40, 60, 80,100	20+20		
		20+20	40, 50, 60, 80,100		140	1
			40, 50, 60, 80,100	20+20		
DC_(n)41DA	DC_(n)41AA	20+20+20	40, 60, 80,100		160	0
			40, 60, 80,100	20+20+20		

		20+20+20	40, 50, 60, 80,100		160	1
			40, 50, 60, 80,100	20+20+20		
DC_(n)71AA	DC_(n)71AA	15	5		20	0
		10	5, 10			
		5	5, 10, 15			
			5	15		
			5, 10	10		
			5, 10, 15	5		
NOTE 1: Void NOTE 2: Void NOTE 3: Void NOTE 4: The channel bandwidths for E-UTRA or NR carrier should be at least supported in one of the BCS indicated in E-UTRA bandwidth combination sets or NR bandwidth combination sets if reported.						

5.3B.1.3 BCS for Intra-band non-contiguous EN-DC

For intra-band non-contiguous EN-DC, an EN-DC configuration is consisting of an E-UTRA band and a corresponding NR band having the same frequency range which supports E-UTRA and NR carriers, where E-UTRA configuration is indicated by using E-UTRA CA bandwidth class as defined in TS 36.101 [4] and NR configuration is indicated by using NR CA bandwidth class as defined in TS 38.101-1 [2].

Requirements for intra-band non-contiguous EN-DC are defined for the EN-DC configurations and bandwidth combination sets specified in Table 5.3B.1.3-1. The EN-DC configurations and bandwidth combination sets in Table 5.3B.1.3-1 also apply to higher order EN-DC combinations that include inter-band and intra-band EN-DC on the downlink and inter-band EN-DC on the uplink. If no BCS is reported in the UE capabilities for an intra-band combination the default is that the UE supports BCS0.

Table 5.3B.1.3-1: EN-DC configurations and bandwidth combination sets defined for intra-band non-contiguous EN-DC

Downlink EN-DC configuration	Uplink EN-DC configurations	E-UTRA – NR configuration / Bandwidth combination set			Maximum aggregated bandwidth (MHz)	Bandwidth combination set
		Component carriers in order of increasing carrier frequency	Channel bandwidths for E-UTRA carrier (MHz)	Channel bandwidths for NR carrier (MHz)		
DC_3A_n3A	DC_3A_n3A ⁽¹⁾		5, 10, 15, 20, 25, 30	5, 10, 15, 20	50	0
DC_41A_n41A	DC_41A_n41A	20	40, 60, 80,100		120	0
			40, 60, 80,100	20		
		20	40, 50, 60, 80,100		120	1
DC_41C_n41A	DC_41A_n41A		40, 50, 60, 80,100	20		
		20+20	40, 60, 80,100		140	0
			40, 60, 80,100	20+20		
DC_41D_n41A	DC_41A_n41A	20+20	40, 50, 60, 80,100		140	1
			40, 50, 60, 80,100	20+20		
		20+20+20	40, 60, 80,100		160	0
			40, 60, 80,100	20+20+20		

		20+20+20	40, 50, 60, 80,100		160	1
			40, 50, 60, 80,100	20+20+20		
NOTE 1: Only single switched UL is supported in Rel.15						
NOTE 2: Void						
NOTE 3: The channel bandwidths for E-UTRA or NR carrier should be at least supported in one of the BCS indicated in E-UTRA bandwidth combination sets or NR bandwidth combination sets if reported						

5.4 Void

5.4A Channel arrangement for CA

The channel arrangement for CA operations in FR1 and FR2 as specified in TS 38.101-1 [2] and TS 38.101-2 [3], respectively.

5.4B Channel arrangement for DC

The channel arrangement for intra-band EN-DC operations in FR1 is specified in TS 36.101 [4] and TS 38.101-1 [2], respectively.

5.4B.1 Channel spacing for intra-band EN-DC carriers

The spacing between carriers will depend on the deployment scenario, the size of the frequency block available and the channel bandwidths. The nominal channel spacing between an E-UTRA carrier and an adjacent NR carrier for intra-band contiguous EN-DC is defined as following:

- For NR operating bands with 100 kHz channel raster,

$$\text{Nominal Channel spacing} = (BW_{\text{E-UTRA_Channel}} + BW_{\text{NR_Channel}})/2$$

- For NR operating bands with 15 kHz channel raster,

- Nominal Channel spacing = $(BW_{\text{E-UTRA_Channel}} + BW_{\text{NR_Channel}})/2 + \{-5\text{kHz}, 0\text{kHz}, 5\text{kHz}\}$ for ΔF_{Raster} equals to 15 kHz

- Nominal Channel spacing = $(BW_{\text{E-UTRA_Channel}} + BW_{\text{NR_Channel}})/2 + \{-10\text{ kHz}, 0\text{ kHz}, 10\text{ kHz}\}$ for ΔF_{Raster} equals to 30 kHz

where $BW_{\text{E-UTRA_Channel}}$ and $BW_{\text{NR_Channel}}$ are the channel bandwidths of the E-UTRA and NR carriers, ΔF_{Raster} is the band dependent channel raster granularity defined in TS38.101-1[2]. The channel spacing can be adjusted depending on the channel raster to optimize performance in a particular deployment scenario.

For intra-band non-contiguous EN-DC the channel spacing between E-UTRA and NR carriers shall be larger than the nominal channel spacing defined in this clause.

5.5 Configuration

5.5A Configuration for CA

5.5A.1 Inter-band CA configurations between FR1 and FR2

Table 5.5A.1-1: Inter-band CA configurations and bandwidth combinations sets between FR1 and FR2 (two bands)

NR CA configuration	Uplink CA configuration	NR Band	SCS (kHz)	5 MHz	10 MHz	15 MHz	20 MHz	40 MHz	50 MHz	60 MHz	80 MHz	90 MHz	100 MHz	200 MHz
CA_n8A-n258A	CA_n8A-n258A	n8	15	Yes	Yes	Yes	Yes							
			30		Yes	Yes	Yes							
			60											
		n258	60						Yes				Yes	Yes
			120						Yes				Yes	Yes
CA_n71A-n257A	-	n71	15	Yes	Yes	Yes	Yes							
			30		Yes	Yes	Yes							
			60											
		n257	60						Yes				Yes	Yes
			120						Yes				Yes	Yes
CA_n77A-n257A	CA_n77A-n257A	n77	15		Yes	Yes	Yes	Yes	Yes					
			30		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
			60		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
		n257	60						Yes				Yes	Yes
			120						Yes				Yes	Yes
CA_n77A-n257D	CA_n77A-n257A	n77	15		Yes	Yes	Yes	Yes	Yes					
			30		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
			60		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
		n257	60						Yes				Yes	Yes
			120						Yes				Yes	Yes
CA_n77A-n257E	CA_n77A-n257A	n77	15	Yes	Yes	Yes	Yes	Yes	Yes					
			30	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
			60	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
		n257		CA_n257E										
				CA_n257E										
CA_n77A-n257F	CA_n77A-n257A	n77	15		Yes	Yes	Yes	Yes	Yes					
			30		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
			60		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
		n257		CA_n257F										
				CA_n257F										
CA_n77C-n257A	CA_n77A-n257A	n77		CA_n77C										
				CA_n77C										
		n257	60						Yes				Yes	Yes
			120						Yes				Yes	Yes
				CA_n257D										
CA_n77C-n257D	CA_n77A-n257A	n77		CA_n77C										
				CA_n77C										
		n257		CA_n257D										
				CA_n257D										
				CA_n257E										
CA_n77C-n257E	CA_n77A-n257A	n77		CA_n77C										
				CA_n77C										
		n257		CA_n257E										
				CA_n257E										
				CA_n257F										
CA_n77C-n257F	CA_n77A-n257A	n77		CA_n77C										
				CA_n77C										
		n257		CA_n257F										
				CA_n257F										
				CA_n257F										
CA_n78A-n257A	CA_n78A-n257A	n78	15		Yes	Yes	Yes	Yes	Yes					
			30		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
			60		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
		n257	60						Yes				Yes	Yes
			120						Yes				Yes	Yes
CA_n78A-n257D	CA_n78A-n257A	n78	15		Yes	Yes	Yes	Yes	Yes					
			30		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
			60		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
		n257	60						Yes				Yes	Yes
			120						Yes				Yes	Yes
CA_n78A-n257E	CA_n78A-n257A	n78	15		Yes	Yes	Yes	Yes	Yes					
			30		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
			60		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
		n257	60						Yes				Yes	Yes
			120						Yes				Yes	Yes
CA_n78A-n257F	CA_n78A-n257A	n78	15		Yes	Yes	Yes	Yes	Yes					
			30		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
			60		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
		n257	60						Yes				Yes	Yes
			120						Yes				Yes	Yes

			30		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
			60		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
		n257		CA_n257E										
CA_n78A-n257F	CA_n78A-n257A	n78	15		Yes	Yes	Yes	Yes	Yes					
			30		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
			60		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
		n257		CA_n257F										
CA_n78C-n257A	CA_n78A-n257A	n78		CA_n78C										
		n257	60						Yes				Yes	Yes
			120						Yes				Yes	Yes
CA_n78C-n257D	CA_n78A-n257A	n78		CA_n78C										
		n257		CA_n257D										
CA_n78C-n257E	CA_n78A-n257A	n78		CA_n78C										
		n257		CA_n257E										
CA_n78C-n257F	CA_n78A-n257A	n78		CA_n78C										
		n257		CA_n257F										
CA_n79A-n257A	CA_n79A-n257A	n79	15					Yes	Yes					
			30					Yes	Yes	Yes	Yes		Yes	
			60					Yes	Yes	Yes	Yes		Yes	
		n257	60						Yes				Yes	Yes
			120						Yes				Yes	Yes
CA_n79A-n257D	CA_n79A-n257A	n79	15					Yes	Yes					
			30					Yes	Yes	Yes	Yes		Yes	
			60					Yes	Yes	Yes	Yes		Yes	
		n257		CA_n257D										
CA_n79A-n257E	CA_n79A-n257A	n79	15					Yes	Yes					
			30					Yes	Yes	Yes	Yes		Yes	
			60					Yes	Yes	Yes	Yes		Yes	
		n257		CA_n257E										
CA_n79A-n257F	CA_n79A-n257A	n79	15					Yes	Yes					
			30					Yes	Yes	Yes	Yes		Yes	
			60					Yes	Yes	Yes	Yes		Yes	
		n257		CA_n257F										
CA_n79C-n257A	CA_n79A-n257A	n79		See CA_n79C in Table 5.5A.1-1 in TS 38.101-1										
		n257	60						Yes				Yes	Yes
			120						Yes				Yes	Yes
CA_n79C-n257D	CA_n79A-n257A	n79		CA_n79C										
		n257		CA_n257D										
CA_n79C-n257E	CA_n79A-n257A	n79		CA_n79C										
		n257		CA_n257E										
CA_n79C-n257F	CA_n79A-n257A	n79		CA_n79C										
		n257		CA_n257F										

NOTE: The CA configurations are given in Table 5.5A.1-1 of either TS 38.101-1 or TS 38.101-2 where unless otherwise stated BCSC

5.5B Configuration for DC

5.5B.1 General

The operating bands and bandwidth classes are specified for operation with EN-DC, NGEN-DC, NE-DC or NR-DC configured. The EN-DC, NGEN-DC or NE-DC band combinations include at least one E-UTRA operating band.

For EN-DC or NE-DC configurations indicated by column "Single Uplink allowed" (e.g., problematic band combinations as defined in TS 38.306 [11]) in tables in this clause the UE may indicate capability of not supporting simultaneous dual and triple uplink operation due to possible intermodulation interference to its own primary downlink channel bandwidth of PCell or PSCell if the intermodulation order is 2 or if the intermodulation order is 3 for the combinations when both operating bands are between 450 MHz – 960 MHz or between 1427 MHz – 2690 MHz.

In the case for EN-DC or NE-DC configurations listed in tables in this clause for which the intermodulation products caused by the dual and triple uplink operation fall into the receive band but do not interfere with its own primary downlink channel bandwidth of PCell or PSCell as defined in Annex I the UE is mandated to operate in dual and triple uplink mode. Single Uplink is also allowed for certain band combinations where intermodulation or reverse intermodulation products could create difficulty for meeting emission requirements.

For EN-DC combinations of order 3 or higher, "Single Uplink allowed" UL configurations captured in Table 5.5B.2-1, Table 5.5B.3-1, and Table 5.5B.4-1 apply. If multiple UL DC configurations are listed for multiple DL DC configurations, valid uplink configurations are such that uplink does not have more carriers than downlink.

Noncontiguous resource allocation and almost contiguous allocation are not applicable for EUTRA or NR carrier part of intraband ENDC configuration.

If the mandatory simultaneous Rx/Tx capability applies for a lower order DC configuration, when the applicable lower order DC configuration is a band pair in a higher order DC configuration, the mandatory simultaneous Rx/Tx capability also applies for the band pair in the higher order DC configuration.

For a higher order EN-DC band combination of which DC_20_n28/ DC_28_n20/ CA_20-28/ CA_n20-n28 is a subset, the frequency range in band n28/28 is restricted for the higher order band combination to 703-733 MHz for the UL and 758-788 MHz for the DL.

For NR inter-band dual connectivity specified in 5.5B.7, the corresponding NR CA configurations in 5.5A.1, i.e., dual uplink inter-band carrier aggregation between FR1 and FR2 with uplink assigned to two NR bands, are applicable to Dual Connectivity.

NOTE 1: Requirements for the dual connectivity configurations are defined in the clause corresponding NR uplink CA between FR1 and FR2 configurations, unless otherwise specified.

5.5B.2 Intra-band contiguous EN-DC

Table 5.5B.2-1: Intra-band contiguous EN-DC configurations

EN-DC configuration	Uplink EN-DC configuration (NOTE 1)	Single UL allowed
DC_(n)41AA ⁵ DC_(n)41CA ⁵ DC_(n)41DA ⁵	DC_(n)41AA	Yes ³
DC_(n)71AA ²	DC_(n)71AA	No ⁴
NOTE 1: Uplink EN-DC configurations are the configurations supported by the present release of specifications. NOTE 2: Requirements in this specification apply for NR SCS of 15 kHz only. NOTE 3: Single UL allowed due to potential emission issues, not self-interference. NOTE 4: For UE(s) supporting dynamic power sharing it is mandatory to do dual simultaneous UL. For UE(s) not supporting dynamic power sharing single UL is allowed. NOTE 5: The minimum requirements only apply for non-simultaneous Tx/Rx between all carriers.		

5.5B.3 Intra-band non-contiguous EN-DC

Table 5.5B.3-1: Intra-band non-contiguous EN-DC configurations

EN-DC configuration	Uplink EN-DC configuration (NOTE 1)	Single UL allowed
DC_3A_n3A	DC_3A_n3A ²	Yes ²
DC_41A_n41A ³ DC_41C_n41A ³ DC_41D_n41A ³	DC_41A_n41A	Yes ⁴
NOTE 1: Uplink EN-DC configurations are the configurations supported by the present release of specifications. NOTE 2: Only single switched UL is supported in Rel.15 NOTE 3: The minimum requirements only apply for non-simultaneous Tx/Rx between all carriers. NOTE 4: Single UL allowed due to potential emission issues, not self-interference.		

5.5B.4 Inter-band EN-DC within FR1

5.5B.4.1 Inter-band EN-DC configurations within FR1 (two bands)

Table 5.5B.4.1-1: Inter-band EN-DC configurations within FR1 (two bands)

EN-DC configuration	Uplink EN-DC configuration (NOTE 1)	Single UL allowed
DC_1A_n28A	DC_1A_n28A	No
DC_1A_n40A	DC_1A_n40A	No
DC_1A_n51A	DC_1A_n51A	No
DC_1A_n77A ⁷ DC_1A_n77C ⁷	DC_1A_n77A	DC_1_n77
DC_1A_n78A ⁷ DC_1A_n78C ⁷	DC_1A_n78A	No
DC_1A_n79A ⁷ DC_1A_n79C ⁷	DC_1A_n79A	No
DC_2A_n5A	DC_2A_n5A	No
DC_2A_n66A	DC_2A_n66A	DC_2_n66

DC_2A_n71A	DC_2A_n71A	No
DC_2A_n78A	DC_2A_n78A	DC_2_n78
DC_3A_n7A	DC_3A_n7A	No
DC_3A_n28A	DC_3A_n28A	No
DC_3A_n40A	DC_3A_n40A	No
DC_3A_n51A	DC_3A_n51A	No
DC_3A_n77A ⁷ DC_3A_n77C ⁷	DC_3A_n77A	DC_3_n77
DC_3A_n78A ⁷ DC_3A_n78C ⁷ DC_3C_n78A ⁷	DC_3A_n78A	DC_3_n78
DC_3A_n79A ⁷ DC_3A_n79C ⁷	DC_3A_n79A	No
DC_5A_n40A	DC_5A_n40A	No
DC_5A_n66A	DC_5A_n66A	DC_5_n66
DC_5A_n78A ⁷	DC_5A_n78A	No
DC_7A-7A_n78A ⁷	DC_7A_n78A	No
DC_7A_n28A	DC_7A_n28A	No
DC_7A_n51A	DC_7A_n51A	No
DC_7A_n78A ⁷	DC_7A_n78A	No
DC_7C_n78A ⁷	DC_7A_n78A	No
DC_8A_n40A ⁷	DC_8A_n40A	No
DC_8A_n77A ⁷	DC_8A_n77A	No
DC_8A_n78A ⁷	DC_8A_n78A	No
DC_8A_n79A ⁷	DC_8A_n79A	No
DC_11A_n77A ⁷	DC_11A_n77A	No
DC_11A_n78A ⁷	DC_11A_n78A	No
DC_11A_n79A ⁷	DC_11A_n79A	No
DC_12A_n5A	DC_12A_n5A	No
DC_12A_n66A	DC_12A_n66A	No
DC_18A_n77A ⁷	DC_18A_n77A	No
DC_18A_n78A ⁷	DC_18A_n78A	No
DC_18A_n79A ⁷	DC_18A_n79A	No
DC_19A_n77A ⁷ DC_19A_n77C ⁷	DC_19A_n77A	No
DC_19A_n78A ⁷ DC_19A_n78C ⁷	DC_19A_n78A	No
DC_19A_n79A ⁷ DC_19A_n79C ⁷	DC_19A_n79A	No
DC_20A_n8A	DC_20A_n8A	DC_20_n8
DC_20A_n28A ^{8,10,11}	DC_20A_n28A	No
DC_20A_n51A	DC_20A_n51A	No
DC_20A_n77A ⁷	DC_20A_n77A	No
DC_20A_n78A ⁷	DC_20A_n78A	No
DC_21A_n77A ⁷ DC_21A_n77C ⁷	DC_21A_n77A	No
DC_21A_n78A ⁷ DC_21A_n78C ⁷	DC_21A_n78A	No
DC_21A_n79A ⁷ DC_21A_n79C ⁷	DC_21A_n79A	No
DC_25A_n41A	DC_25A_n41A	No
DC_26A_n41A	DC_26A_n41A	No
DC_26A_n77A ⁷	DC_26A_n77A	No

DC_26A_n78A ⁷	DC_26A_n78A	No
DC_26A_n79A ⁷	DC_26A_n79A	No
DC_28A_n51A	DC_28A_n51A	No
DC_28A_n77A ⁷ DC_28A_n77C ⁷	DC_28A_n77A	No
DC_28A_n78A ⁷ DC_28A_n78C ⁷	DC_28A_n78A	No
DC_28A_n79A ⁷ DC_28A_n79C ⁷	DC_28A_n79A	No
DC_30A_n5A	DC_30A_n5A	No
DC_30A_n66A	DC_30A_n66A	No
DC_38A_n78A ⁷	N/A	No
DC_39A_n78A ^{5,7}	DC_39A_n78A	No
DC_39A_n79A ⁷	DC_39A_n79A	No
DC_40A_n77A	N/A	No
DC_41A_n77A DC_41C_n77A	DC_41A_n77A	No
DC_41A_n78A DC_41C_n78A	DC_41A_n78A	No
DC_41A_n79A ^{6,7} DC_41C_n79A ^{6,7}	DC_41A_n79A	No
DC_42A_n51A	DC_42A_n51A	No
DC_42A_n77A ^{3,4,9} DC_42A_n77C ^{3,4,9} DC_42C_n77A ^{3,4,9} DC_42C_n77C ^{3,4,9} DC_42D_n77A ^{3,4,9} DC_42E_n77A ^{3,4,9}	N/A	N/A
DC_42A_n78A ^{3,4,9} DC_42A_n78C ^{3,4,9} DC_42C_n78A ^{3,4,9} DC_42C_n78C ^{3,4,9} DC_42D_n78A ^{3,4,9} DC_42E_n78A ^{3,4,9}	N/A	N/A
DC_42A_n79A ^{9,15} DC_42A_n79C ^{9,15} DC_42C_n79A ^{9,15} DC_42C_n79C ^{9,15} DC_42D_n79A ^{9,15} DC_42E_n79A ^{9,15}	N/A	N/A
DC_46A_n78A ² DC_46C_n78A ² DC_46D_n78A ² DC_46E_n78A ²	N/A	N/A
DC_66A_n5A	DC_66A_n5A	DC_66_n5
DC_66A_n71A	DC_66A_n71A	No
DC_66A_n78A	DC_66A_n78A	No
NOTE 1: Uplink EN-DC configurations are the configurations supported by the present release of specifications. NOTE 2: Restricted to E-UTRA operation when inter-band carrier aggregation is configured. The downlink operating band for Band 46 is paired with the uplink operating band (external E-UTRA band) of the carrier aggregation configuration that is supporting the configured Pcell. NOTE 3: The minimum requirements apply only when there is non-simultaneous Tx/Rx operation between E-UTRA and NR carriers. This restriction applies also for these carriers when applicable EN-DC configuration is part of a higher order EN-DC configuration. NOTE 4: The minimum requirements for intra-band non-contiguous EN-DC apply. When UE capability <i>interBandContiguousMRDC</i> is indicated, the minimum requirements for intra-band-contiguous EN-DC also should be met in addition to intra-band non-contiguous EN-DC. The intra-band requirements also apply for these carriers when		

	applicable EN-DC configuration is a subset of a higher order EN-DC configuration.
NOTE 5:	The frequency range above 3600 MHz for Band n78 is not used in this combination.
NOTE 6:	The frequency range below 2506 MHz for Band 41 is not used in this combination.
NOTE 7:	Applicable for UE supporting inter-band EN-DC with mandatory simultaneous Rx/Tx capability.
NOTE 8:	The frequency range in band n28 / 28 is restricted for this band combination to 703 - 733 MHz for the UL and 758-788 MHz for the DL. This restriction also apply for any band combinations when DC_20_n28/ DC_28_n20/ CA_20-28/ CA_n20-n28 is a subset of a higher order band combination.
NOTE 9:	The combination is not used alone as fall back mode of other band combinations in which UL in Band 42 is not used.
NOTE 10:	The minimum requirements apply for DL carriers with a maximum power spectral density imbalance of [6] dB. The power spectral density imbalance condition also applies for these carriers when applicable EN-DC configuration is a subset of a higher order EN-DC configuration
NOTE 11:	The minimum requirements apply for synchronized DL carriers with a maximum receive time difference ≤ 3 usec. The requirements also apply for these carriers when applicable EN-DC configuration is a subset of a higher order EN-DC configuration
NOTE 12:	Void.
NOTE 13:	Void.
NOTE 14:	Void.
NOTE 15:	Simultaneous Rx/Tx capability does not apply for UEs supporting band 42 with a n77 implementation only. Same restrictions are applied to related higher order configurations.

5.5B.4.2 Inter-band EN-DC configurations within FR1 (three bands)

Table 5.5B.4.2-1: Inter-band EN-DC configurations within FR1 (three bands)

EN-DC configuration	Uplink EN-DC configuration (NOTE 1)
DC_1A-3A_n28A	DC_1A_n28A DC_3A_n28A
DC_1A-3A_n77A ⁵ DC_1A-3A_n77C ⁵	DC_1A_n77A DC_3A_n77A
DC_1A-3A_n78A ⁵ DC_1A-3A_n78C ⁵ DC_1A-3C_n78A ⁵	DC_1A_n78A DC_3A_n78A
DC_1A-3A_n79A ⁵ DC_1A-3A_n79C ⁵	DC_1A_n79A DC_3A_n79A
DC_1A-5A_n78A ⁵	DC_1A_n78A DC_5A_n78A
DC_1A-7A_n28A ⁵	DC_1A_n28A DC_7A_n28A
DC_1A-7A_n78A ⁵	DC_1A_n78A DC_7A_n78A
DC_1A-7A-7A_n78A ⁵	DC_1A_n78A DC_7A_n78A
DC_1A-8A_n78A ⁵	DC_1A_n78A DC_8A_n78A
DC_1A-18A_n77A ⁵	DC_1A_n77A DC_18A_n77A
DC_1A-18A_n78A ⁵	DC_1A_n78A DC_18A_n78A
DC_1A-18A_n79A	DC_1A_n79A DC_18A_n79A
DC_1A-19A_n77A ⁵ DC_1A-19A_n77C ⁵	DC_1A_n77A DC_19A_n77A
DC_1A-19A_n78A ⁵ DC_1A-19A_n78C ⁵	DC_1A_n78A DC_19A_n78A
DC_1A-19A_n79A ⁵ DC_1A-19A_n79C ⁵	DC_1A_n79A DC_19A_n79A

DC_1A-20A_n28A ⁶	DC_1A_n28A DC_20A_n28A
DC_1A-20A_n78A ⁵	DC_1A_n78A DC_20A_n78A
DC_1A-21A_n77A ⁵ DC_1A-21A_n77C ⁵	DC_1A_n77A DC_21A_n77A
DC_1A-21A_n78A ⁵ DC_1A-21A_n78C ⁵	DC_1A_n78A DC_21A_n78A
DC_1A-21A_n79A ⁵ DC_1A-21A_n79C ⁵	DC_1A_n79A DC_21A_n79A
DC_1A-28A_n77A ⁵ DC_1A-28A_n77C ⁵	DC_1A_n77A DC_28A_n77A
DC_1A-28A_n78A ⁵ DC_1A-28A_n78C ⁵	DC_1A_n78A DC_28A_n78A
DC_1A_n28A-n78A ⁵	DC_1A_n28A DC_1A_n78A
DC_1A-28A_n79A DC_1A-28A_n79C	DC_1A_n79A DC_28A_n79A
DC_1A-41A_n77A DC_1A-41C_n77A	DC_1A_n77A DC_41A_n77A
DC_1A-41A_n78A DC_1A-41C_n78A	DC_1A_n78A DC_41A_n78A
DC_1A-41C_n79A	DC_1A_n79A
DC_1A-42A_n77A DC_1A-42A_n77C DC_1A-42C_n77A DC_1A-42C_n77C DC_1A-42D_n77A DC_1A-42E_n77A	DC_1A_n77A
DC_1A-42A_n78A DC_1A-42A_n78C DC_1A-42C_n78A DC_1A-42C_n78C DC_1A-42D_n78A DC_1A-42E_n78A	DC_1A_n78A
DC_1A-42A_n79A DC_1A-42A_n79C DC_1A-42C_n79A DC_1A-42C_n79C DC_1A-42D_n79A DC_1A-42E_n79A	DC_1A_n79A
DC_1A_n77A-n79A ⁷	DC_1A_n77A DC_1A_n79A
DC_1A_n78A-n79A ⁸	DC_1A_n78A DC_1A_n79A
DC_1A_SUL_n78A-n84A ⁵	DC_1A_n78A, DC_1A_n84A_ULSUP-TDM_n78A,
DC_2A-5A_n66A	DC_2A_n66A DC_5A_n66A
DC_2A-12A_n66A	DC_2A_n66A DC_12A_n66A
DC_2A-30A_n66A	DC_2A_n66A DC_30A_n66A
DC_2A-66A_n71A	DC_2A_n71A DC_66A_n71A
DC_2A-(n)71AA	DC_2A_n71A DC_(n)71AA

DC_3A_n3A-n77A	DC_3A_n77A DC_3A_n3A ²
DC_3A_n3A-n78A	DC_3A_n78A DC_3A_n3A ²
DC_3A-5A_n78A ⁵	DC_3A_n78A DC_5A_n78A
DC_3A-7A_n28A	DC_3A_n28A DC_7A_n28A
DC_3A-7A_n78A ⁵ DC_3C-7A_n78A ⁵	DC_3A_n78A DC_7A_n78A
DC_3A-7C_n78A ⁵ DC_3C-7C_n78A ⁵	DC_3A_n78A DC_7A_n78A
DC_3A-7A-7A_n78A ⁵	DC_3A_n78A DC_7A_n78A
DC_3A-8A_n78A	DC_3A_n78A DC_8A_n78A
DC_3A-19A_n77A ⁵ DC_3A-19A_n77C ⁵	DC_3A_n77A DC_19A_n77A
DC_3A-19A_n78A ⁵ DC_3A-19A_n78C ⁵	DC_3A_n78A DC_19A_n78A
DC_3A-19A_n79A ⁵ DC_3A-19A_n79C ⁵	DC_3A_n79A DC_19A_n79A
DC_3A-20A_n28A ^{5,6}	DC_3A_n28A DC_20A_n28A
DC_3A-20A_n78A ⁵ DC_3C-20A_n78A ⁵	DC_3A_n78A DC_20A_n78A
DC_3A-21A_n77A ⁵ DC_3A-21A_n77C ⁵	DC_3A_n77A DC_21A_n77A
DC_3A-21A_n78A ⁵ DC_3A-21A_n78C ⁵	DC_3A_n78A DC_21A_n78A
DC_3A-21A_n79A ⁵ DC_3A-21A_n79C ⁵	DC_3A_n79A DC_21A_n79A
DC_3A-28A_n77A DC_3A-28A_n77C	DC_3A_n77A DC_28A_n77A
DC_3A-28A_n78A ⁵ DC_3A-28A_n78C ⁵	DC_3A_n78A DC_28A_n78A
DC_3A_n28A-n78A ⁵	DC_3A_n28A DC_3A_n78A
DC_3A-28A_n79A DC_3A-28A_n79C	DC_3A_n79A DC_28A_n79A
DC_3A-38A_n78A	DC_3A_n78A
DC_3A-41A_n78A	DC_3A_n78A DC_41A_n78A
DC_3A-42A_n77A DC_3A-42A_n77C DC_3A-42C_n77A DC_3A-42C_n77C DC_3A-42D_n77A DC_3A-42E_n77A	DC_3A_n77A
DC_3A-42A_n78A DC_3A-42A_n78C DC_3A-42C_n78A DC_3A-42C_n78C DC_3A-42D_n78A DC_3A-42E_n78A	DC_3A_n78A
DC_3A-42A_n79A DC_3A-42A_n79C DC_3A-42C_n79A DC_3A-42C_n79C DC_3A-42D_n79A DC_3A-42E_n79A	DC_3A_n79A

DC_3A_n77A-n79A ⁷	DC_3A_n77A DC_3A_n79A
DC_3A_n78A-n79A ⁸	DC_3A_n78A DC_3A_n79A
DC_3A_SUL_n78A-n80A ⁵	DC_3A_n78A DC_3A_n80A_ULSUP-TDM_n78A
DC_3A_SUL_n78A-n82A ⁵	DC_3A_n78A DC_3A_n82A
DC_3A_SUL_n79A-n80A ⁵	DC_3A_n79A, DC_3A_n80A_ULSUP-TDM_n79A
DC_5A-7A_n78A	DC_5A_n78A DC_7A_n78A
DC_5A-7A-7A_n78A	DC_5A_n78A DC_7A_n78A
DC_5A-30A_n66A	DC_5A_n66A DC_30A_n66A
DC_7A-20A_n28A ⁶	DC_7A_n28A DC_20A_n28A
DC_7A-20A_n78A ⁵	DC_7A_n78A DC_20A_n78A
DC_7A-28A_n78A ⁵	DC_7A_n78A DC_28A_n78A
DC_7C-28A_n78A ⁵	DC_7A_n78A DC_28A_n78A
DC_7A_n28A-n78A ⁵	DC_7A_n28A, DC_7A_n78A
DC_7A-46A_n78A ³ DC_7A-46C_n78A ³ DC_7A-46D_n78A ³ DC_7A-46E_n78A ³	DC_7A_n78A
DC_8A_SUL_n78A-n81A ⁵	DC_8A_n78A, DC_8A_n81A_ULSUP-TDM_n78A,
DC_8A_SUL_n79A-n81A ⁵	DC_8A_n79A, DC_8A_n81A_ULSUP-TDM_n79A,
DC_12A-30A_n66A	DC_12A_n66A DC_30A_n66A
DC_18A-28A_n77A ⁵	DC_18A_n77A DC_28A_n77A
DC_18A-28A_n78A ⁵	DC_18A_n78A DC_28A_n78A
DC_18A-28A_n79A ⁵	DC_18A_n79A DC_28A_n79A
DC_19A-21A_n78A ⁵ DC_19A-21A_n78C ⁵	DC_19A_n78A DC_21A_n78A
DC_19A-21A_n79A ⁵ DC_19A-21A_n79C ⁵	DC_19A_n79A DC_21A_n79A
DC_19A-21A_n77A ⁵ DC_19A-21A_n77C ⁵	DC_19A_n77A DC_21A_n77A
DC_19A-42A_n77A DC_19A-42A_n77C DC_19A-42C_n77A DC_19A-42C_n77C	DC_19A_n77A
DC_19A-42A_n78A DC_19A-42A_n78C DC_19A-42C_n78A DC_19A-42C_n78C	DC_19A_n78A
DC_19A-42A_n79A DC_19A-42A_n79C DC_19A-42C_n79A	DC_19A_n79A

DC_19A-42C_n79C	
DC_19A_n77A-n79A ⁷	DC_19A_n77A DC_19A_n79A
DC_19A_n78A-n79A ⁸	DC_19A_n78A DC_19A_n79A
DC_20A_n8A-n75A ⁶	DC_20A_n8A
DC_20A_n28A-n75A ⁶	DC_20A_n28A
DC_20A_n28A-n78A ^{5,6}	DC_20A_n28A DC_20A_n78A
DC_20A_n75A-n78A ⁵	DC_20A_n78A
DC_20A_n76A-n78A ⁵	DC_20A_n78A
DC_20A_SUL_n78A-n82A ⁵	DC_20A_n78A DC_20A_n82A_ULSUP-TDM_n78A
DC_20A_SUL_n78A-n83A ⁵	DC_20A_n78A DC_20A_n83A
DC_21A-28A_n77A DC_21A-28A_n77C	DC_21A_n77A DC_28A_n77A
DC_21A-28A_n78A DC_21A-28A_n78C	DC_21A_n78A DC_28A_n78A
DC_21A-28A_n79A DC_21A-28A_n79C	DC_21A_n79A DC_28A_n79A
DC_21A-42A_n77A DC_21A-42A_n77C DC_21A-42C_n77A DC_21A-42C_n77C	DC_21A_n77A
DC_21A-42A_n78A DC_21A-42A_n78C DC_21A-42C_n78A DC_21A-42C_n78C	DC_21A_n78A
DC_21A-42A_n79A DC_21A-42A_n79C DC_21A-42C_n79A DC_21A-42C_n79C	DC_21A_n79A
DC_21A_n77A-n79A ⁷	DC_21A_n77A DC_21A_n79A
DC_21A_n78A-n79A ⁸	DC_21A_n78A DC_21A_n79A
DC_28A-42A_n77A DC_28A-42A_n77C DC_28A-42C_n77A	DC_28A_n77A
DC_28A-42A_n78A DC_28A-42A_n78C DC_28A-42C_n78A	DC_28A_n78A
DC_28A-42A_n79A DC_28A-42A_n79C DC_28A-42C_n79A	DC_28A_n79A
DC_28A_SUL_n78A-n83A ⁵	DC_28A_n78A, DC_28A_n83A_ULSUP-TDM_n78A,
DC_41A-42A_n77A DC_41A-42C_n77A DC_41C-42A_n77A DC_41C-42C_n77A	DC_41A_n77A
DC_41A-42A_n78A DC_41A-42C_n78A DC_41C-42A_n78A DC_41C-42C_n78A	DC_41A_n78A
DC_41A-42A_n79A DC_41A-42C_n79A DC_41C-42A_n79A	DC_41A_n79A

DC_41C-42C_n79A	
DC_66A_(n)71AA	DC_66A_n71A DC_(n)71AA
DC_66A_SUL_n78A-n86A ⁵	DC_66A_n78A DC_66A_n86A_ULSUP-TDM_n78A
NOTE 1: Uplink EN-DC configurations are the configurations supported by the present release of specifications. NOTE 2: Only single switched UL is supported in Rel-15 NOTE 3: Restricted to E-UTRA operation when inter-band carrier aggregation is configured. The downlink operating band for Band 46 is paired with the uplink operating band (external E-UTRA band) of the carrier aggregation configuration that is supporting the configured Pcell. NOTE 4: If a UE is configured with both NR UL and NR SUL carriers in a cell, the switching time between NR UL carrier and NR SUL carrier can be up to 140us and placed in SUL resources. NOTE 5: Applicable for UE supporting inter-band EN-DC with mandatory simultaneous Rx/Tx capability NOTE 6: The frequency range in band n28 is restricted for this band combination to 703-733 MHz for the UL and 758-788 MHz for the DL. NOTE 7: The minimum requirements apply only when there is non-simultaneous Rx/Tx operation between n77-n79 NR carriers. This restriction applies also for these carriers when applicable EN-DC configuration is part of a higher order configuration. NOTE 8: For UEs supporting band n77, the minimum requirements apply only when there is non-simultaneous Rx/Tx operation between n78-n79 NR carriers. This restriction applies also for these carriers when applicable EN-DC configuration is part of a higher order configuration.	

5.5B.4.3 Inter-band EN-DC configurations within FR1 (four bands)

Table 5.5B.4.3-1: Inter-band EN-DC configurations within FR1 (four bands)

EN-DC configuration	Uplink EN-DC configuration (NOTE 1)
DC_1A-3A-5A_n78A ²	DC_1A_n78A DC_3A_n78A DC_5A_n78A
DC_1A-3A-7A_n28A	DC_1A_n28A DC_3A_n28A DC_7A_n28A
DC_1A-3A-7A_n78A ² DC_1A-3C-7A_n78A ²	DC_1A_n78A DC_3A_n78A DC_7A_n78A
DC_1A-3A-7A-7A_n78A ²	DC_1A_n78A DC_3A_n78A DC_7A_n78A
DC_1A-3A-8A_n78A ²	DC_1A_n78A DC_3A_n78A DC_8A_n78A
DC_1A-3A-19A_n77A ² DC_1A-3A-19A_n77C ²	DC_1A_n77A DC_3A_n77A DC_19A_n77A
DC_1A-3A-19A_n78A ² DC_1A-3A-19A_n78C ²	DC_1A_n78A DC_3A_n78A DC_19A_n78A
DC_1A-3A-19A_n79A ² DC_1A-3A-19A_n79C ²	DC_1A_n79A DC_3A_n79A DC_19A_n79A
DC_1A-3A-20A_n28A ³	DC_1A_n28A DC_3A_n28A DC_20A_n28A
DC_1A-3A-20A_n78A ²	DC_1A_n78A DC_3A_n78A DC_20A_n78A

DC_1A-3A-21A_n77A ² DC_1A-3A-21A_n77C ²	DC_1A_n77A DC_3A_n77A DC_21A_n77A
DC_1A-3A-21A_n78A ² DC_1A-3A-21A_n78C ²	DC_1A_n78A DC_3A_n78A DC_21A_n78A
DC_1A-3A-21A_n79A ² DC_1A-3A-21A_n79C ²	DC_1A_n79A DC_3A_n79A DC_21A_n79A
DC_1A-3A-28A_n77A ²	DC_1A_n77A DC_3A_n77A DC_28A_n77A
DC_1A-3A-28A_n78A ²	DC_1A_n78A DC_3A_n78A DC_28A_n78A
DC_1A-3A-28A_n79A ²	DC_1A_n79A DC_3A_n79A DC_28A_n79A
DC_1A-3A_n28A-n78A ²	DC_1A_n28A DC_1A_n78A DC_3A_n28A DC_3A_n78A
DC_1A-3A-42A_n77A DC_1A-3A-42A_n77C DC_1A-3A-42C_n77A DC_1A-3A-42C_n77C	DC_1A_n77A DC_3A_n77A
DC_1A-3A-42A_n78A DC_1A-3A-42A_n78C DC_1A-3A-42C_n78A DC_1A-3A-42C_n78C	DC_1A_n78A DC_3A_n78A
DC_1A-3A-42A_n79A DC_1A-3A-42A_n79C DC_1A-3A-42C_n79A DC_1A-3A-42C_n79C	DC_1A_n79A DC_3A_n79A
DC_1A-5A-7A_n78A	DC_1A_n78A DC_5A_n78A DC_7A_n78A
DC_1A-5A-7A-7A_n78A	DC_1A_n78A DC_5A_n78A DC_7A_n78A
DC_1A-7A-20A_n28A ³	DC_1A_n28A DC_7A_n28A DC_20A_n28A
DC_1A-7A-20A_n78A ²	DC_1A_n78A DC_7A_n78A DC_20A_n78A
DC_1A-7A_n28A-n78A ²	DC_1A_n28A DC_1A_n78A DC_7A_n28A DC_7A_n78A
DC_1A-18A-28A_n77A	DC_1A_n77A DC_18A_n77A DC_28A_n77A
DC_1A-18A-28A_n78A	DC_1A_n78A DC_18A_n78A DC_28A_n78A
DC_1A-18A-28A_n79A ²	DC_1A_n79A DC_18A_n79A DC_28A_n79A
DC_1A-19A-21A_n77A DC_1A-19A-21A_n77C	DC_1A_n77A DC_19A_n77A DC_21A_n77A

DC_1A-19A-21A_n78A DC_1A-19A-21A_n78C	DC_1A_n78A DC_19A_n78A DC_21A_n78A
DC_1A-19A-21A_n79A DC_1A-19A-21A_n79C	DC_1A_n79A DC_19A_n79A DC_21A_n79A
DC_1A-19A-42A_n77A DC_1A-19A-42A_n77C DC_1A-19A-42C_n77A DC_1A-19A-42C_n77C	DC_1A_n77A DC_19A_n77A
DC_1A-19A-42A_n78A DC_1A-19A-42A_n78C DC_1A-19A-42C_n78A DC_1A-19A-42C_n78C	DC_1A_n78A DC_19A_n78A
DC_1A-19A-42A_n79A DC_1A-19A-42A_n79C DC_1A-19A-42C_n79A DC_1A-19A-42C_n79C	DC_1A_n79A DC_19A_n79A
DC_1A-20A_n28A-n78A ^{2,3}	DC_1A_n28A DC_1A_n78A DC_20A_n28A DC_20A_n78A
DC_1A-21A-28A_n77A ²	DC_1A_n77A DC_21A_n77A DC_28A_n77A
DC_1A-21A-28A_n78A ²	DC_1A_n78A DC_21A_n78A DC_28A_n78A
DC_1A-21A-28A_n79A ²	DC_1A_n79A DC_21A_n79A DC_28A_n79A
DC_1A-21A-42A_n77A DC_1A-21A-42A_n77C DC_1A-21A-42C_n77A DC_1A-21A-42C_n77C	DC_1A_n77A DC_21A_n77A
DC_1A-21A-42A_n78A DC_1A-21A-42A_n78C DC_1A-21A-42C_n78A DC_1A-21A-42C_n78C	DC_1A_n78A DC_21A_n78A
DC_1A-21A-42A_n79A DC_1A-21A-42A_n79C DC_1A-21A-42C_n79A DC_1A-21A-42C_n79C	DC_1A_n79A DC_21A_n79A
DC_1A-28A-42A_n77A DC_1A-28A-42C_n77A	DC_1A_n77A DC_28A_n77A
DC_1A-28A-42A_n78A DC_1A-28A-42C_n78A	DC_1A_n78A DC_28A_n78A
DC_1A-28A-42A_n79A DC_1A-28A-42C_n79A	DC_1A_n79A DC_28A_n79A
DC_1A-41A-42A_n77A DC_1A-41A-42C_n77A DC_1A-41C-42A_n77A DC_1A-41C-42C_n77A	DC_1A_n77A DC_41A_n77A
DC_1A-41A-42A_n78A DC_1A-41A-42C_n78A DC_1A-41C-42A_n78A DC_1A-41C-42C_n78A	DC_1A_n78A DC_41A_n78A
DC_1A-41A-42A_n79A DC_1A-41A-42C_n79A DC_1A-41C-42A_n79A DC_1A-41C-42C_n79A	DC_1A_n79A DC_41A_n79A
DC_2A-66A-(n)71AA	DC_2A_n71A DC_66A_n71A DC_(n)71AA

DC_3A-5A-7A_n78A	DC_3A_n78A DC_5A_n78A DC_7A_n78A
DC_3A-5A-7A-7A_n78A	DC_3A_n78A DC_5A_n78A DC_7A_n78A
DC_3A-7A-20A_n28A ³	DC_3A_n28A DC_7A_n28A DC_20A_n28A
DC_3A-7A-20A_n78A ²	DC_3A_n78A DC_20A_n78A DC_7A_n78A
DC_3A-7A-28A_n78A ² DC_3A-7C-28A_n78A ²	DC_3A_n78A DC_7A_n78A DC_28A_n78A
DC_3A-7A_n28A-n78A ²	DC_3A_n28A DC_3A_n78A DC_7A_n28A DC_7A_n78A
DC_3A-19A-21A_n77A ² DC_3A-19A-21A_n77C ²	DC_3A_n77A DC_19A_n77A DC_21A_n77A
DC_3A-19A-21A_n78A ² DC_3A-19A-21A_n78C ²	DC_3A_n78A DC_19A_n78A DC_21A_n78A
DC_3A-19A-21A_n79A ² DC_3A-19A-21A_n79C ²	DC_3A_n79A DC_19A_n79A DC_21A_n79A
DC_3A-19A-42A_n77A DC_3A-19A-42A_n77C DC_3A-19A-42C_n77A DC_3A-19A-42C_n77C	DC_3A_n77A DC_19A_n77A
DC_3A-19A-42A_n78A DC_3A-19A-42A_n78C DC_3A-19A-42C_n78A DC_3A-19A-42C_n78C	DC_3A_n78A DC_19A_n78A
DC_3A-19A-42A_n79A ² DC_3A-19A-42A_n79C ² DC_3A-19A-42C_n79A ² DC_3A-19A-42C_n79C ²	DC_3A_n79A DC_19A_n79A
DC_3A-20A_n28A-n78A ^{2,3}	DC_3A_n28A DC_3A_n78A DC_20A_n28A DC_20A_n78A
DC_3A-21A-42A_n77A DC_3A-21A-42A_n77C DC_3A-21A-42C_n77A DC_3A-21A-42C_n77C	DC_3A_n77A DC_21A_n77A
DC_3A-21A-42A_n78A DC_3A-21A-42A_n78C DC_3A-21A-42C_n78A DC_3A-21A-42C_n78C	DC_3A_n78A DC_21A_n78A
DC_3A-21A-42A_n79A DC_3A-21A-42A_n79C DC_3A-21A-42C_n79A DC_3A-21A-42C_n79C	DC_3A_n79A DC_21A_n79A
DC_3A-28A-42A_n77A DC_3A-28A-42C_n77A	DC_3A_n77A DC_28A_n77A
DC_3A-28A-42A_n78A DC_3A-28A-42C_n78A	DC_3A_n78A DC_28A_n78A
DC_3A-28A-42A_n79A DC_3A-28A-42C_n79A	DC_3A_n79A DC_28A_n79A

DC_7A-20A_n28A-n78A ^{2,3}	DC_7A_n28A DC_7A_n78A DC_20A_n28A DC_20A_n78A
DC_19A-21A-42A_n77A DC_19A-21A-42A_n77C DC_19A-21A-42C_n77A DC_19A-21A-42C_n77C	DC_19A_n77A DC_21A_n77A
DC_19A-21A-42A_n78A DC_19A-21A-42A_n78C DC_19A-21A-42C_n78A DC_19A-21A-42C_n78C	DC_19A_n78A DC_21A_n78A
DC_19A-21A-42A_n79A DC_19A-21A-42A_n79C DC_19A-21A-42C_n79A DC_19A-21A-42C_n79C	DC_19A_n79A DC_21A_n79A
DC_21A-28A-42A_n77A DC_21A-28A-42C_n77A	DC_21A_n77A DC_28A_n77A
DC_21A-28A-42A_n78A DC_21A-28A-42C_n78A	DC_21A_n78A DC_28A_n78A
DC_21A-28A-42A_n79A DC_21A-28A-42C_n79A	DC_21A_n79A DC_28A_n79A
NOTE 1: Uplink EN-DC configurations are the configurations supported by the present release of specifications. NOTE 2: Applicable for UE supporting inter-band EN-DC with mandatory simultaneous Rx/Tx capability NOTE 3: The frequency range in band n28 is restricted for this band combination to 703-733 MHz for the UL and 758-788 MHz for the DL.	

5.5B.4.4 Inter-band EN-DC configurations within FR1 (five bands)

Table 5.5B.4.4-1: Inter-band EN-DC configurations within FR1 (five bands)

EN-DC configuration	Uplink EN-DC configuration (NOTE 1)
DC_1A-3A-5A-7A_n78A	DC_1A_n78A DC_3A_n78A DC_5A_n78A DC_7A_n78A
DC_1A-3A-5A-7A-7A_n78A	DC_1A_n78A DC_3A_n78A DC_5A_n78A DC_7A_n78A
DC_1A-3A-7A-20A_n28A ³	DC_1A_n28A DC_3A_n28A DC_7A_n28A DC_20A_n28A
DC_1A-3A-7A-20A_n78A ²	DC_1A_n78A DC_3A_n78A DC_7A_n78A DC_20A_n78A
DC_1A-3A-7A_n28A-n78A ²	DC_1A_n28A DC_1A_n78A DC_3A_n28A DC_3A_n78A DC_7A_n28A DC_7A_n78A
DC_1A-3A-19A-21A_n77A ² DC_1A-3A-19A-21A_n77C ²	DC_1A_n77A DC_3A_n77A DC_19A_n77A DC_21A_n77A

DC_1A-3A-19A-21A_n78A ² DC_1A-3A-19A-21A_n78C ²	DC_1A_n78A DC_3A_n78A DC_19A_n78A DC_21A_n78A
DC_1A-3A-19A-21A_n79A ² DC_1A-3A-19A-21A_n79C ²	DC_1A_n79A DC_3A_n79A DC_19A_n79A DC_21A_n79A
DC_1A-3A-19A-42A_n77A DC_1A-3A-19A-42A_n77C DC_1A-3A-19A-42C_n77A DC_1A-3A-19A-42C_n77C	DC_1A_n77A DC_3A_n77A DC_19A_n77A
DC_1A-3A-19A-42A_n78A DC_1A-3A-19A-42A_n78C DC_1A-3A-19A-42C_n78A DC_1A-3A-19A-42C_n78C	DC_1A_n78A DC_3A_n78A DC_19A_n78A
DC_1A-3A-19A-42A_n79A DC_1A-3A-19A-42A_n79C DC_1A-3A-19A-42C_n79A DC_1A-3A-19A-42C_n79C	DC_1A_n79A DC_3A_n79A DC_19A_n79A
DC_1A-3A-20A_n28A-n78A ^{2,3}	DC_1A_n28A DC_1A_n78A DC_3A_n28A DC_3A_n78A DC_20A_n28A DC_20A_n78A
DC_1A-3A-21A-42A_n77A DC_1A-3A-21A-42A_n77C DC_1A-3A-21A-42C_n77A DC_1A-3A-21A-42C_n77C	DC_1A_n77A DC_3A_n77A DC_21A_n77A
DC_1A-3A-21A-42A_n78A DC_1A-3A-21A-42A_n78C DC_1A-3A-21A-42C_n78A DC_1A-3A-21A-42C_n78C	DC_1A_n78A DC_3A_n78A DC_21A_n78A
DC_1A-3A-21A-42A_n79A DC_1A-3A-21A-42A_n79C DC_1A-3A-21A-42C_n79A DC_1A-3A-21A-42C_n79C	DC_1A_n79A DC_3A_n79A DC_21A_n79A
DC_1A-3A-28A-42A_n77A DC_1A-3A-28A-42C_n77A	DC_1A_n77A DC_3A_n77A DC_28A_n77A
DC_1A-3A-28A-42A_n78A DC_1A-3A-28A-42C_n78A	DC_1A_n78A DC_3A_n78A DC_28A_n78A
DC_1A-3A-28A-42A_n79A DC_1A-3A-28A-42C_n79A	DC_1A_n79A DC_3A_n79A DC_28A_n79A
DC_1A-7A-20A_n28A-n78A ^{2,3}	DC_1A_n28A DC_1A_n78A DC_7A_n28A DC_7A_n78A DC_20A_n28A DC_20A_n78A
DC_1A-19A-21A-42A_n77A DC_1A-19A-21A-42A_n77C DC_1A-19A-21A-42C_n77A DC_1A-19A-21A-42C_n77C	DC_1A_n77A DC_19A_n77A DC_21A_n77A
DC_1A-19A-21A-42A_n78A DC_1A-19A-21A-42A_n78C DC_1A-19A-21A-42C_n78A DC_1A-19A-21A-42C_n78C	DC_1A_n78A DC_19A_n78A DC_21A_n78A
DC_1A-19A-21A-42A_n79A DC_1A-19A-21A-42A_n79C DC_1A-19A-21A-42C_n79A DC_1A-19A-21A-42C_n79C	DC_1A_n79A DC_19A_n79A DC_21A_n79A

DC_1A-21A-28A-42A_n77A DC_1A-21A-28A-42C_n77A	DC_1A_n77A DC_21A_n77A DC_28A_n77A
DC_1A-21A-28A-42A_n78A DC_1A-21A-28A-42C_n78A	DC_1A_n78A DC_21A_n78A DC_28A_n78A
DC_1A-21A-28A-42A_n79A DC_1A-21A-28A-42C_n79A	DC_1A_n79A DC_21A_n79A DC_28A_n79A
DC_3A-7A-20A_n28A-n78A ^{2,3}	DC_3A_n28A DC_3A_n78A DC_7A_n28A DC_7A_n78A DC_20A_n28A DC_20A_n78A
NOTE 1: Uplink EN-DC configurations are the configurations supported by the present release of specifications. NOTE 2: Applicable for UE supporting inter-band EN-DC with mandatory simultaneous Rx/Tx capability NOTE 3: The frequency range in band n28 is restricted for this band combination to 703-733 MHz for the UL and 758-788 MHz for the DL	

5.5B.4.5 Inter-band EN-DC configurations within FR1 (six bands)

Table 5.5B.4.5-1: Inter-band EN-DC configurations within FR1 (six bands)

EN-DC configuration	Uplink EN-DC configuration (NOTE 1)
DC_1A-3A-7A-20A_n28A-n78A ^{2,3}	DC_1A_n28A DC_1A_n78A DC_3A_n28A DC_3A_n78A DC_7A_n28A DC_7A_n78A DC_20A_n28A DC_20A_n78A
NOTE 1: Uplink EN-DC configurations are the configurations supported by the present release of specifications. NOTE 2: Applicable for UE supporting inter-band EN-DC with mandatory simultaneous Rx/Tx capability NOTE 3: The frequency range in band n28 is restricted for this band combination to 703-733 MHz for the UL and 758-788 MHz for the DL	

5.5B.4a Inter-band NE-DC within FR1

5.5B.4a.1 Inter-band NE-DC configurations within FR1 (two bands)

Table 5.5B.4a.1-1: Inter-band NE-DC configurations within FR1 (two bands)

NE-DC configuration	Uplink NE-DC configuration (NOTE 1)	Single UL allowed
DC_n1A_28A	DC_n1A_28A	No
NOTE 1: Uplink NE-DC configurations are the configurations supported by the present release of specifications.		

5.5B.5 Inter-band EN-DC including FR2

5.5B.5.1 Inter-band EN-DC configurations including FR2 (two bands)

Table 5.5B.5.1-1: Inter-band EN-DC configurations including FR2 (two bands)

EN-DC configuration	Uplink EN-DC configuration (NOTE 1)
DC_1A_n257A DC_1A_n257D DC_1A_n257E DC_1A_n257F	DC_1A_n257A
DC_2A_n257A DC_2C_n257A	DC_2A_n257A
DC_2A_n257(2A)	DC_2A_n257A
DC_2A-2A_n257A	DC_2A_n257A
DC_2A_n260A DC_2A_n260G DC_2A_n260H DC_2A_n260I DC_2A_n260J DC_2A_n260K DC_2A_n260L DC_2A_n260M DC_2C_n260A	DC_2A_n260A
DC_2A_n260(2A)	DC_2A_n260
DC_2A-2A_n260A DC_2A-2A_n260G DC_2A-2A_n260H DC_2A-2A_n260I DC_2A-2A_n260J DC_2A-2A_n260K DC_2A-2A_n260L DC_2A-2A_n260M	DC_2A_n260A
DC_3A_n257A DC_3A_n257D DC_3A_n257E DC_3A_n257F	DC_3A_n257A
DC_3A_n258A	DC_3A_n258A
DC_5A_n257A DC_5B_n257A	DC_5A_n257A DC_5B_n257A
DC_5A-5A_n257A	DC_5A_n257A
DC_5A_n260A DC_5A_n260B DC_5A_n260C DC_5A_n260D DC_5A_n260E DC_5A_n260F DC_5A_n260G DC_5A_n260H DC_5A_n260I DC_5A_n260J DC_5A_n260K DC_5A_n260L DC_5A_n260M DC_5A_n260O DC_5A_n260P DC_5A_n260Q DC_5B_n260A	DC_5A_n260A DC_5B_n260A
DC_5A_n260(2A) DC_5A_n260(3A) DC_5A_n260(4A) DC_5A_n260(A-I) DC_5A_n260(D-G) DC_5A_n260(D-H) DC_5A_n260(D-I) DC_5A_n260(D-O)	DC_5A_n260A

DC_5A_n260(D-P) DC_5A_n260(D-Q) DC_5A_n260(E-O) DC_5A_n260(E-P) DC_5A_n260(E-Q) DC_5A_n260(G-I)	
DC_5A-5A_n260A	DC_5A_n260A
DC_5A_n261A DC_5A_n261B DC_5A_n261C DC_5A_n261D DC_5A_n261E DC_5A_n261F DC_5A_n261G DC_5A_n261H DC_5A_n261I DC_5A_n261J DC_5A_n261K DC_5A_n261L DC_5A_n261M DC_5A_n261O DC_5A_n261P DC_5A_n261Q	DC_5A_n261A
DC_5A_n261(2A) DC_5A_n261(3A) DC_5A_n261(4A) DC_5A_n261(D-G) DC_5A_n261(D-H) DC_5A_n261(D-I) DC_5A_n261(D-O) DC_5A_n261(D-P) DC_5A_n261(D-Q) DC_5A_n261(E-O) DC_5A_n261(E-P) DC_5A_n261(E-Q)	DC_5A_n261A
DC_7A_n257A	DC_7A_n257A
DC_7A-7A_n257A	DC_7A_n257A
DC_7A_n258A	DC_7A_n258A
DC_8A_n257A	DC_8A_n257A
DC_8A_n258A	DC_8A_n258A
DC_11A_n257A	DC_11A_n257A
DC_12A_n260A DC_12A_n260G DC_12A_n260H DC_12A_n260I DC_12A_n260J DC_12A_n260K DC_12A_n260L DC_12A_n260M	DC_12A_n260A
DC_12A_n260(A-I) DC_12A_n260(G-I)	DC_12A_n260A
DC_13A_n257A	DC_13A_n257A
DC_13A_n260A	DC_13A_n260A
DC_18A_n257A	DC_18A_n257A
DC_19A_n257A DC_19A_n257D DC_19A_n257E DC_19A_n257F	DC_19A_n257A
DC_20A_n258A	DC_20A_n258A
DC_21A_n257A DC_21A_n257D DC_21A_n257E DC_21A_n257F	DC_21A_n257A
DC_26A_n257A	DC_26A_n257A

DC_28A_n257A DC_28A_n257D DC_28A_n257E DC_28A_n257F	DC_28A_n257A
DC_28A_n258A	DC_28A_n258A
DC_30A_n260A DC_30A_n260G DC_30A_n260H DC_30A_n260I DC_30A_n260J DC_30A_n260K DC_30A_n260L DC_30A_n260M	DC_30A_n260A
DC_30A_n260(A-I) DC_30A_n260(G-I)	DC_30A_n260A
DC_39A_n258A	DC_39A_n258A
DC_41A_n257A DC_41C_n257A	DC_41A_n257A DC_41C_n257A
DC_41A_n258A	DC_41A_n258A
DC_42A_n257A DC_42C_n257A DC_42A_n257D DC_42A_n257E DC_42A_n257F DC_42C_n257D DC_42C_n257E DC_42C_n257F DC_42D_n257A DC_42E_n257A	DC_42A_n257A DC_42C_n257A
DC_48A_n257A DC_48C_n257A	DC_48A_n257A DC_48C_n257A
DC_48A-48A_n257A	DC_48A_n257A
DC_48A_n260A DC_48C_n260A	DC_48A_n260A DC_48C_n260A
DC_48A-48A_n260A	DC_48A_n260A
DC_66A_n257A DC_66A_n257G DC_66A_n257H DC_66A_n257I DC_66A_n257J DC_66A_n257K DC_66A_n257L DC_66A_n257M DC_66C_n257A	DC_66A_n257A
DC_66A_n257(2A)	DC_66A_n257A
DC_66A-66A_n257A	DC_66A_n257A
DC_66A_n260A DC_66A_n260D DC_66A_n260E DC_66A_n260F DC_66A_n260G DC_66A_n260H DC_66A_n260I DC_66A_n260J DC_66A_n260K DC_66A_n260L DC_66A_n260M DC_66A_n260O DC_66A_n260P DC_66A_n260Q	DC_66A_n260A
DC_66A_n260(2A) DC_66A_n260(3A) DC_66A_n260(4A) DC_66A_n260(A-I) DC_66A_n260(D-G)	DC_66A_n260A

DC_66A_n260(D-H) DC_66A_n260(D-I) DC_66A_n260(D-O) DC_66A_n260(D-P) DC_66A_n260(D-Q) DC_66A_n260(E-O) DC_66A_n260(E-P) DC_66A_n260(E-Q) DC_66A_n260(G-I)	
DC_66A-66A_n260A DC_66A-66A_n260G DC_66A-66A_n260H DC_66A-66A_n260I DC_66A-66A_n260J DC_66A-66A_n260K DC_66A-66A_n260L DC_66A-66A_n260M	DC_66A_n260A
DC_66A_n261A DC_66A_n261D DC_66A_n261E DC_66A_n261F DC_66A_n261G DC_66A_n261H DC_66A_n261I DC_66A_n261J DC_66A_n261K DC_66A_n261L DC_66A_n261M DC_66A_n261O DC_66A_n261P DC_66A_n261Q	DC_66A_n261A
DC_66A_n261(2A) DC_66A_n261(3A) DC_66A_n261(4A) DC_66A_n261(D-G) DC_66A_n261(D-H) DC_66A_n261(D-I) DC_66A_n261(D-O) DC_66A_n261(D-P) DC_66A_n261(D-Q) DC_66A_n261(E-O) DC_66A_n261(E-P) DC_66A_n261(E-Q)	DC_66A_n261A
NOTE 1: Uplink EN-DC configurations are the configurations supported by the present release of specifications. NOTE 2: Applicable for UE supporting inter-band EN-DC with mandatory simultaneous Rx/Tx capability for all of the above combinations	

5.5B.5.2 Inter-band EN-DC configurations including FR2 (three bands)

Table 5.5B.5.2-1: Inter-band EN-DC configurations including FR2 (three bands)

EN-DC configuration	Uplink EN-DC configuration (NOTE 1)
DC_1A-3A_n257A ² DC_1A-3A_n257D ² DC_1A-3A_n257E ² DC_1A-3A_n257F ²	DC_1A_n257A DC_3A_n257A
DC_1A-5A_n257A ²	DC_1A_n257A DC_5A_n257A
DC_1A-7A_n257A ²	DC_1A_n257A DC_7A_n257A
DC_1A-7A-7A_n257A ²	DC_1A_n257A DC_7A_n257A
DC_1A-8A_n257A ²	DC_1A_n257A DC_8A_n257A

DC_1A-18A_n257A ²	DC_1A_n257A DC_18A_n257A
DC_1A-19A_n257A ² DC_1A-19A_n257D ² DC_1A-19A_n257E ² DC_1A-19A_n257F ²	DC_1A_n257A DC_19A_n257A
DC_1A-21A_n257A ² DC_1A-21A_n257D ² DC_1A-21A_n257E ² DC_1A-21A_n257F ²	DC_1A_n257A DC_21A_n257A
DC_1A-28A_n257A ² DC_1A-28A_n257D ² DC_1A-28A_n257E ² DC_1A-28A_n257F ²	DC_1A_n257A DC_28A_n257A
DC_1A-41A_n257A DC_1A-41C_n257A	DC_1A_n257A DC_41A_n257A DC_41C_n257A
DC_1A-42A_n257A DC_1A-42A_n257D DC_1A-42A_n257E DC_1A-42A_n257F DC_1A-42C_n257A DC_1A-42C_n257D DC_1A-42C_n257E DC_1A-42C_n257F DC_1A-42D_n257A DC_1A-42E_n257A	DC_1A_n257A DC_42A_n257A
DC_2A-5A_n257A ²	DC_2A_n257A DC_5A_n257A
DC_2A-5A_n260A DC_2A-5A_n260G DC_2A-5A_n260H DC_2A-5A_n260I DC_2A-5A_n260J DC_2A-5A_n260K DC_2A-5A_n260L DC_2A-5A_n260M	DC_2A_n260A DC_5A_n260A
DC_2A-12A_n260A DC_2A-12A_n260G DC_2A-12A_n260H DC_2A-12A_n260I DC_2A-12A_n260J DC_2A-12A_n260K DC_2A-12A_n260L DC_2A-12A_n260M	DC_2A_n260A DC_12A_n260A
DC_2A-13A_n257A ²	DC_2A_n257A DC_13A_n257A
DC_2A-13A_n260A ²	DC_2A_n260A DC_13A_n260A
DC_2A-30A_n260A DC_2A-30A_n260G DC_2A-30A_n260H DC_2A-30A_n260I DC_2A-30A_n260J DC_2A-30A_n260K DC_2A-30A_n260L DC_2A-30A_n260M	DC_2A_n260A DC_30A_n260A
DC_2A-66A_n257A ²	DC_2A_n257A DC_66A_n257A
DC_2A-66A_n260A DC_2A-66A_n260G DC_2A-66A_n260H DC_2A-66A_n260I DC_2A-66A_n260J DC_2A-66A_n260K DC_2A-66A_n260L DC_2A-66A_n260M	DC_2A_n260A DC_66A_n260A

DC_3A-5A_n257A ²	DC_3A_n257A DC_5A_n257A
DC_3A-7A_n257A ²	DC_3A_n257A DC_7A_n257A
DC_3A-7A-7A_n257A ²	DC_3A_n257A DC_7A_n257A
DC_3A-19A_n257A ² DC_3A-19A_n257D ² DC_3A-19A_n257E ² DC_3A-19A_n257F ²	DC_3A_n257A DC_19A_n257A
DC_3A-21A_n257A ² DC_3A-21A_n257D ² DC_3A-21A_n257E ² DC_3A-21A_n257F ²	DC_3A_n257A DC_21A_n257A
DC_3A-28A_n257A ² DC_3A-28A_n257D ² DC_3A-28A_n257E ² DC_3A-28A_n257F ²	DC_3A_n257A DC_28A_n257A
DC_3A-41A_n257A	DC_3A_n257A DC_41A_n257A
DC_3A-42A_n257A ² DC_3A-42A_n257D ² DC_3A-42A_n257E ² DC_3A-42A_n257F ² DC_3A-42C_n257A ² DC_3A-42C_n257D ² DC_3A-42C_n257E ² DC_3A-42C_n257F ² DC_3A-42D_n257A ² DC_3A-42E_n257A ²	DC_3A_n257A DC_42A_n257A
DC_5A-7A_n257A ²	DC_5A_n257A DC_7A_n257A
DC_5A-7A-7A_n257A	DC_5A_n257A DC_7A_n257A
DC_5A-30A_n260A DC_5A-30A_n260G DC_5A-30A_n260H DC_5A-30A_n260I DC_5A-30A_n260J DC_5A-30A_n260K DC_5A-30A_n260L DC_5A-30A_n260M	DC_5A_n260A DC_30A_n260A
DC_5A-66A_n257A	DC_5A_n257A DC_66A_n257A
DC_5A-66A_n260A DC_5A-66A_n260G DC_5A-66A_n260H DC_5A-66A_n260I DC_5A-66A_n260J DC_5A-66A_n260K DC_5A-66A_n260L DC_5A-66A_n260M	DC_5A_n260A DC_66A_n260A
DC_12A-30A_n260A DC_12A-30A_n260G DC_12A-30A_n260H DC_12A-30A_n260I DC_12A-30A_n260J DC_12A-30A_n260K DC_12A-30A_n260L DC_12A-30A_n260M	DC_12A_n260A DC_30A_n260A
DC_12A-66A_n260A DC_12A-66A_n260G DC_12A-66A_n260H DC_12A-66A_n260I DC_12A-66A_n260J DC_12A-66A_n260K	DC_12A_n260A DC_66A_n260A

DC_12A-66A_n260L DC_12A-66A_n260M	
DC_13A-66A_n257A ²	DC_13A_n257A DC_66A_n257A
DC_13A-66A_n260A ²	DC_13A_n260A DC_66A_n260A
DC_18A-28A_n257A ²	DC_18A_n257A DC_28A_n257A
DC_19A-21A_n257A ² DC_19A-21A_n257D ² DC_19A-21A_n257E ² DC_19A-21A_n257F ²	DC_19A_n257A DC_21A_n257A
DC_19A-42A_n257A ² DC_19A-42A_n257D ² DC_19A-42A_n257E ² DC_19A-42A_n257F ² DC_19A-42C_n257A ²	DC_19A_n257A DC_42A_n257A
DC_21A-28A_n257A ² DC_21A-28A_n257D ² DC_21A-28A_n257E ² DC_21A-28A_n257F ²	DC_21A_n257A DC_28A_n257A
DC_21A-42A_n257A ² DC_21A-42A_n257D ² DC_21A-42A_n257E ² DC_21A-42A_n257F ² DC_21A-42C_n257A ²	DC_21A_n257A DC_42A_n257A
DC_28A-42C_n257A ² DC_28A-42A_n257A ²	DC_28A_n257A DC_42A_n257A
DC_30A-66A_n260A DC_30A-66A_n260G DC_30A-66A_n260H DC_30A-66A_n260I DC_30A-66A_n260J DC_30A-66A_n260K DC_30A-66A_n260L DC_30A-66A_n260M	DC_30A_n260A DC_66A_n260A
DC_41A-42A_n257A DC_41A-42C_n257A DC_41C-42A_n257A DC_41C-42C_n257A	DC_41A_n257A DC_42A_n257A
NOTE 1: Uplink EN-DC configurations are the configurations supported by the present release of specifications. NOTE 2: Applicable for UE supporting inter-band EN-DC with mandatory simultaneous Rx/Tx capability	

5.5B.5.3 Inter-band EN-DC configurations including FR2 (four bands)

Table 5.5B.5.3-1: Inter-band EN-DC configurations including FR2 (four bands)

EN-DC configuration	Uplink EN-DC configuration (NOTE 1)
DC_1A-3A-5A_n257A ²	DC_1A_n257A DC_3A_n257A DC_5A_n257A
DC_1A-3A-7A_n257A ²	DC_1A_n257A DC_3A_n257A DC_7A_n257A
DC_1A-3A-7A-7A_n257A	DC_1A_n257A DC_3A_n257A DC_7A_n257A
DC_1A-3A-19A_n257A ²	DC_1A_n257A DC_3A_n257A DC_19A_n257A

DC_1A-3A-21A_n257A ²	DC_1A_n257A DC_3A_n257A DC_21A_n257A
DC_1A-3A-28A_n257A ²	DC_1A_n257A DC_3A_n257A DC_28A_n257A
DC_1A-3A-42A_n257A DC_1A-3A-42C_n257A DC_1A-3A-42C_n257D DC_1A-3A-42C_n257E DC_1A-3A-42C_n257F	DC_1A_n257A DC_3A_n257A DC_42A_n257A
DC_1A-5A-7A_n257A ²	DC_1A_n257A DC_5A_n257A DC_7A_n257A
DC_1A-5A-7A-7A_n257A	DC_1A_n257A DC_5A_n257A DC_7A_n257A
DC_1A-18A-28A_n257A ²	DC_1A_n257A DC_18A_n257A DC_28A_n257A
DC_1A-19A-21A_n257A DC_1A-19A-21A_n257D DC_1A-19A-21A_n257E DC_1A-19A-21A_n257F	DC_1A_n257A DC_19A_n257A DC_21A_n257A
DC_1A-19A-42A_n257A DC_1A-19A-42C_n257A DC_1A-19A-42C_n257D DC_1A-19A-42C_n257E DC_1A-19A-42C_n257F	DC_1A_n257A DC_19A_n257A DC_42A_n257A
DC_1A-21A-28A_n257A ²	DC_1A_n257A DC_21A_n257A DC_28A_n257A
DC_1A-21A-42A_n257A DC_1A-21A-42C_n257A DC_1A-21A-42C_n257D DC_1A-21A-42C_n257E DC_1A-21A-42C_n257F	DC_1A_n257A DC_21A_n257A DC_42A_n257A
DC_1A-28A-42A_n257A DC_1A-28A-42C_n257A	DC_1A_n257A DC_28A_n257A DC_42A_n257A
DC_1A-41A-42A_n257A DC_1A-41A-42C_n257A DC_1A-41C-42A_n257A DC_1A-41C-42C_n257A	DC_1A_n257A DC_41A_n257A DC_42A_n257A
DC_3A-5A-7A_n257A ²	DC_3A_n257A DC_5A_n257A DC_7A_n257A
DC_3A-5A-7A-7A_n257A ²	DC_3A_n257A DC_5A_n257A DC_7A_n257A
DC_3A-19A-21A_n257A ²	DC_3A_n257A DC_19A_n257A DC_21A_n257A
DC_3A-19A-42A_n257A DC_3A-19A-42C_n257A DC_3A-19A-42C_n257D DC_3A-19A-42C_n257E DC_3A-19A-42C_n257F	DC_3A_n257A DC_19A_n257A DC_42A_n257A
DC_3A-21A-42A_n257A DC_3A-21A-42C_n257A DC_3A-21A-42C_n257D DC_3A-21A-42C_n257E DC_3A-21A-42C_n257F	DC_3A_n257A DC_21A_n257A DC_42A_n257A
DC_3A-28A-42A_n257A DC_3A-28A-42C_n257A	DC_3A_n257A DC_28A_n257A DC_42A_n257A

DC_19A-21A-42A_n257A ² DC_19A-21A-42C_n257A ² DC_19A-21A-42C_n257D ² DC_19A-21A-42C_n257E ² DC_19A-21A-42C_n257F ²	DC_19A_n257A DC_21A_n257A DC_42A_n257A
DC_21A-28A-42A_n257A ² DC_21A-28A-42C_n257A ²	DC_21A_n257A DC_28A_n257A DC_42A_n257A
NOTE 1: Uplink EN-DC configurations are the configurations supported by the present release of specifications. NOTE 2: Applicable for UE supporting inter-band EN-DC with mandatory simultaneous Rx/Tx capability	

5.5B.5.4 Inter-band EN-DC configurations including FR2 (five bands)

Table 5.5B.5.4-1: Inter-band EN-DC configurations including FR2 (five bands)

EN-DC configuration	Uplink EN-DC configuration (NOTE 1)
DC_1A-3A-5A-7A_n257A	DC_1A_n257A DC_3A_n257A DC_5A_n257A DC_7A_n257A
DC_1A-3A-5A-7A-7A_n257A	DC_1A_n257A DC_3A_n257A DC_5A_n257A DC_7A_n257A
DC_1A-3A-19A-21A_n257A DC_1A-3A-19A-21A_n257D DC_1A-3A-19A-21A_n257E DC_1A-3A-19A-21A_n257F	DC_1A_n257A DC_3A_n257A DC_19A_n257A DC_21A_n257A
DC_1A-3A-19A-42A_n257A DC_1A-3A-19A-42A_n257D DC_1A-3A-19A-42A_n257E DC_1A-3A-19A-42A_n257F DC_1A-3A-19A-42C_n257A DC_1A-3A-19A-42C_n257D DC_1A-3A-19A-42C_n257E DC_1A-3A-19A-42C_n257F	DC_1A_n257A DC_3A_n257A DC_19A_n257A DC_42A_n257A
DC_1A-3A-21A-42A_n257A DC_1A-3A-21A-42C_n257A DC_1A-3A-21A-42C_n257D DC_1A-3A-21A-42C_n257E DC_1A-3A-21A-42C_n257F	DC_1A_n257A DC_3A_n257A DC_21A_n257A DC_42A_n257A
DC_1A-3A-28A-42A_n257A DC_1A-3A-28A-42C_n257A	DC_1A_n257A DC_3A_n257A DC_28A_n257A DC_42A_n257A
DC_1A-19A-21A-42A_n257A DC_1A-19A-21A-42A_n257D DC_1A-19A-21A-42A_n257E DC_1A-19A-21A-42A_n257F DC_1A-19A-21A-42C_n257A DC_1A-19A-21A-42C_n257D DC_1A-19A-21A-42C_n257E DC_1A-19A-21A-42C_n257F	DC_1A_n257A DC_19A_n257A DC_21A_n257A DC_42A_n257A
DC_1A-19A-28A-42C_n257A	DC_1A_n257A DC_19A_n257A DC_28A_n257A DC_42A_n257A
DC_1A-21A-28A-42A_n257A	DC_1A_n257A DC_21A_n257A DC_28A_n257A DC_42A_n257A
NOTE 1: Uplink EN-DC configurations are the configurations supported by the present release of specifications. NOTE 2: Void	

5.5B.5.5 Void

5.5B.6 Inter-band EN-DC including FR1 and FR2

5.5B.6.1 Void

5.5B.6.2 Inter-band EN-DC configurations including FR1 and FR2 (three bands)

Table 5.5B.6.2-1: Inter-band EN-DC configurations including FR1 and FR2 (three bands)

EN-DC configuration	Uplink EN-DC configuration (NOTE 1)
DC_1A_n77A-n257A DC_1A_n77A-n257D DC_1A_n77A-n257E DC_1A_n77A-n257F DC_1A_n77C-n257A DC_1A_n77C-n257D DC_1A_n77C-n257E DC_1A_n77C-n257F	DC_1A_n77A DC_1A_n257A DC_1A_n77A-n257A
DC_1A_n78A-n257A DC_1A_n78A-n257D DC_1A_n78A-n257E DC_1A_n78A-n257F DC_1A_n78C-n257A DC_1A_n78C-n257D DC_1A_n78C-n257E DC_1A_n78C-n257F	DC_1A_n78A DC_1A_n257A DC_1A_n78A-n257A
DC_1A_n79A-n257A DC_1A_n79A-n257D DC_1A_n79A-n257E DC_1A_n79A-n257F DC_1A_n79C-n257A DC_1A_n79C-n257D DC_1A_n79C-n257E DC_1A_n79C-n257F	DC_1A_n79A DC_1A_n257A DC_1A_n79A-n257A
DC_3A_n77A-n257A DC_3A_n77A-n257D DC_3A_n77A-n257E DC_3A_n77A-n257F DC_3A_n77C-n257A DC_3A_n77C-n257D DC_3A_n77C-n257E DC_3A_n77C-n257F	DC_3A_n77A DC_3A_n257A DC_3A_n77A-n257A
DC_3A_n78A-n257A DC_3A_n78A-n257D DC_3A_n78A-n257E DC_3A_n78A-n257F DC_3A_n78C-n257A DC_3A_n78C-n257D DC_3A_n78C-n257E DC_3A_n78C-n257F	DC_3A_n78A DC_3A_n257A DC_3A_n78A-n257A
DC_3A_n79A-n257A DC_3A_n79A-n257D DC_3A_n79A-n257E DC_3A_n79A-n257F DC_3A_n79C-n257A DC_3A_n79C-n257D DC_3A_n79C-n257E DC_3A_n79C-n257F	DC_3A_n79A DC_3A_n257A DC_3A_n79A-n257A
DC_5A_n78A-n257A ²	DC_5A_n78A DC_5A_n257A
DC_7A_n78A-n257A	DC_7A_n78A DC_7A_n257A

DC_7A-7A_n78A-n257A	DC_7A_n78A DC_7A_n257A DC_7A_n78A-n257A
DC_19A_n77A-n257A DC_19A_n77A-n257D DC_19A_n77A-n257E DC_19A_n77A-n257F DC_19A_n77C-n257A DC_19A_n77C-n257D DC_19A_n77C-n257E DC_19A_n77C-n257F	DC_19A_n77A DC_19A_n257A DC_19A_n77A-n257A
DC_19A_n78A-n257A DC_19A_n78A-n257D DC_19A_n78A-n257E DC_19A_n78A-n257F DC_19A_n78C-n257A DC_19A_n78C-n257D DC_19A_n78C-n257E DC_19A_n78C-n257F	DC_19A_n78A DC_19A_n257A DC_19A_n78A-n257A
DC_19A_n79A-n257A DC_19A_n79A-n257D DC_19A_n79A-n257E DC_19A_n79A-n257F DC_19A_n79C-n257A DC_19A_n79C-n257D DC_19A_n79C-n257E DC_19A_n79C-n257F	DC_19A_n79A DC_19A_n257A DC_19A_n79A-n257A
DC_21A_n77A-n257A	DC_21A_n77A DC_21A_n257A
DC_21A_n78A-n257A	DC_21A_n78A DC_21A_n257A
DC_21A_n79A-n257A	DC_21A_n79A DC_21A_n257A
NOTE 1: Uplink EN-DC configurations are the configurations supported by the present release of specifications. NOTE 2: Applicable for UE supporting inter-band EN-DC with mandatory simultaneous Rx/Tx capability	

5.5B.6.3 Inter-band EN-DC configurations including FR1 and FR2 (four bands)

Table 5.5B.6.3-1: Inter-band EN-DC configurations including FR1 and FR2 (four bands)

EN-DC configuration	Uplink EN-DC configuration (NOTE 1)
DC_1A-3A_n78A-n257A	DC_1A_n78A DC_1A_n257A DC_3A_n78A DC_3A_n257A
DC_1A-5A_n78A-n257A	DC_1A_n78A DC_1A_n257A DC_5A_n78A DC_5A_n257A
DC_1A-7A_n78A-n257A	DC_1A_n78A DC_1A_n257A DC_7A_n78A DC_7A_n257A
DC_1A-7A-7A_n78A-n257A	DC_1A_n78A DC_1A_n257A DC_7A_n78A DC_7A_n257A
DC_3A-5A_n78A-n257A	DC_3A_n78A DC_3A_n257A DC_5A_n78A

	DC_5A_n257A
DC_3A-7A_n78A-n257A	DC_3A_n78A DC_3A_n257A DC_7A_n78A DC_7A_n257A
DC_3A-7A-7A_n78A-n257A	DC_3A_n78A DC_3A_n257A DC_7A_n78A DC_7A_n257A
DC_5A-7A_n78A-n257A	DC_5A_n78A DC_5A_n257A DC_7A_n78A DC_7A_n257A
DC_5A-7A-7A_n78A-n257A	DC_5A_n78A DC_5A_n257A DC_7A_n78A DC_7A_n257A
NOTE 1: Uplink EN-DC configurations are the configurations supported by the present release of specifications.	

5.5B.6.4 Inter-band EN-DC configurations including FR1 and FR2 (five bands)

Table 5.5B.6.4-1: Inter-band EN-DC configurations including FR1 and FR2 (five bands)

EN-DC configuration	Uplink EN-DC configuration (NOTE 1)
DC_1A-3A-5A_n78A-n257A	DC_1A_n78A DC_1A_n257A DC_3A_n78A DC_3A_n257A DC_5A_n78A DC_5A_n257A
DC_1A-3A-7A_n78A-n257A	DC_1A_n78A DC_1A_n257A DC_3A_n78A DC_3A_n257A DC_7A_n78A DC_7A_n257A
DC_1A-3A-7A-7A_n78A-n257A	DC_1A_n78A DC_1A_n257A DC_3A_n78A DC_3A_n257A DC_7A_n78A DC_7A_n257A
DC_1A-5A-7A_n78A-n257A	DC_1A_n78A DC_1A_n257A DC_5A_n78A DC_5A_n257A DC_7A_n78A DC_7A_n257A
DC_1A-5A-7A-7A_n78A-n257A	DC_1A_n78A DC_1A_n257A DC_5A_n78A DC_5A_n257A DC_7A_n78A DC_7A_n257A
DC_3A-5A-7A_n78A-n257A	DC_3A_n78A DC_3A_n257A DC_5A_n78A DC_5A_n257A DC_7A_n78A DC_7A_n257A
DC_3A-5A-7A-7A_n78A-n257A	DC_3A_n78A DC_3A_n257A DC_5A_n78A

	DC_5A_n257A DC_7A_n78A DC_7A_n257A
NOTE 1: Uplink EN-DC configurations are the configurations supported by the present release of specifications.	

5.5B.6.5 Inter-band EN-DC configurations including FR1 and FR2 (six bands)

Table 5.5B.6.5-1: Inter-band EN-DC configurations including FR1 and FR2 (six bands)

EN-DC configuration	Uplink EN-DC configuration (NOTE 1)
DC_1A-3A-5A-7A_n78A-n257A	DC_1A_n78A DC_1A_n257A DC_3A_n78A DC_3A_n257A DC_5A_n78A DC_5A_n257A DC_7A_n78A DC_7A_n257A
NOTE 1: Uplink EN-DC configurations are the configurations supported by the present release of specifications.	

5.5B.7 Inter-band NR-DC between FR1 and FR2

5.5B.7.0 General

The configurations and bandwidth combination sets for the FR1-FR2 NR-DC combinations in the following sub-section are defined in the tables for FR1-FR2 carrier aggregation in section 5.5A.1.

5.5B.7.1 Inter-band NR-DC configurations between FR1 and FR2 (two bands)

Table 5.5B.7-1: Inter-band NR-DC configurations between FR1 and FR2 (two bands)

Downlink NR DC configuration	Uplink NR DC configuration
DC_n77A-n257A DC_n77A-n257D DC_n77A-n257E DC_n77A-n257F DC_n77A-n257G DC_n77A-n257H DC_n77A-n257I DC_n77A-n257J DC_n77A-n257K DC_n77A-n257L DC_n77A-n257M DC_n77C-n257A DC_n77C-n257D DC_n77C-n257E DC_n77C-n257F	DC_n77A-n257A
DC_n78A-n257A DC_n78A-n257D DC_n78A-n257E DC_n78A-n257F DC_n78A-n257G DC_n78A-n257H DC_n78A-n257I DC_n78A-n257J	DC_n78A-n257A

DC_n78A-n257K DC_n78A-n257L DC_n78A-n257M DC_n78C-n257A DC_n78C-n257D DC_n78C-n257E DC_n78C-n257F	
DC_n79A-n257A DC_n79A-n257D DC_n79A-n257E DC_n79A-n257F DC_n79A-n257G DC_n79A-n257H DC_n79A-n257I DC_n79A-n257J DC_n79A-n257K DC_n79A-n257L DC_n79A-n257M DC_n79C-n257A DC_n79C-n257D DC_n79C-n257E DC_n79C-n257F	DC_n79A-n257A